

3D-Printing for Supply Chain Resilience

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Abstract. Problem definition: Most strategies for mitigating supply chain disruptions require upfront investments for each sourced component, making them impractical for large, disaggregated supply chains. We study 3D printing (3DP) as a flexible backup resource that addresses disruptions across multiple suppliers and components. Although 3DP has traditionally been viewed as ill-suited for resilience due to high per-unit costs and limited capacity, we explore when and why it may nonetheless be cost-effective and how it should be deployed. **Methodology/results:** We formulate an optimization model to decide which components to back up, how much 3DP capacity to procure, and how to allocate it in real time to unmet demand. The resulting mixed-binary stochastic optimization problem is computationally challenging, even for moderately sized supply chains. We develop a novel framework combining Taylor-series-based supermodular approximations with first-order stochastic optimization to compute high-quality solutions, and use these bounds to motivate simple, interpretable, portfolio-level screening features. The analysis yields structural insights, including partitions of backup strategies and bounds on ordering and capacity decisions. Finally, we conduct a case study calibrated to bill-of-lading data from toy manufacturer Mattel. With current cost estimates, 3DP offers modest savings relative to traditional resilience strategies despite higher unit costs, but substantially reduces demand shortfalls, particularly in larger systems with weakly correlated disruptions. **Managerial implications:** Our findings provide actionable guidance on introducing 3DP into an existing portfolio of resilience strategies. We characterize when even a small investment in 3DP capacity yields cost savings, providing a basis for limited-scale piloting. Moreover, our derived portfolio-level screening features predict backup-switching decisions more effectively than marginal attributes alone. This challenges the conventional component-by-component approach to dedicated backups: managers should view 3DP backup design as a portfolio-design problem, assessing whether each candidate component diversifies the existing portfolio or congests scarce capacity in the same disruption scenarios.

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1. Introduction

Supply chain disruptions are part of normal operations (Simchi-Levi et al. 2015). Poorly managed disruptions can be extremely costly. Some estimates suggest losses of up to 45% of annual earnings over a decade (McKinsey & Company 2021), prompting many firms to invest in resilience. Indeed, in a recent survey, 97% of respondents reported implementing at least one resilience measure following COVID-19 (McKinsey & Company 2023).

Common supply chain resilience strategies include inventory buffering (Simchi-Levi et al. 2015, 2017), dual sourcing or siting (Tomlin 2006, Dada et al. 2007), and reserving capacity at backup suppliers (Yang et al. 2009, Demirel et al. 2017). These strategies are typically applied in a dedicated, “per component” or “per supplier” manner, requiring separate upfront investments. Dedicated approaches can be difficult to implement for components that are critical to final production, yet low-volume or low-margin (for the supplier), since suppliers may be unwilling to invest in dual sites or reserve capacity. Such components are common in large, disaggregated supply chains. For example, Simchi-Levi et al. (2015) report that Ford’s supply chain involves over \$35 billion in parts sourced from 14,000 Tier 1 suppliers across 4,400 sites. Many of these parts have high “performance impact,” meaning they are critical to production, yet they generate low margins for suppliers.

In sum, firms are left with two unpalatable options: eat the cost of comprehensive, dedicated backup, or wear the risk of leaving some components unprotected.

In this paper, we explore a third, unconventional option: leveraging 3D Printing (3DP), also known as additive manufacturing, as a pooled, capacitated, flexible backup resource. By building items layer by layer from scratch, a single printer can produce a wide range of components from plastic polymers, metals, or composites—without incurring the component-specific fixed costs of traditional manufacturing techniques such as injection molding. Most importantly, 3DP allows firms to defer the choice of *which* components to print until after supplier disruptions are realized.

We stress that our goal is *not* to revisit whether this flexibility is valuable—a point now well established in both manufacturing (Jordan and Graves 1995, Simchi-Levi and Wei 2012) and service systems (Bassamboo et al. 2012, Tsitsiklis and Xu 2013)—but, rather to understand how flexible backup can be operationalized in large, disaggregated supply chains:

- (i) Which components should be backed up using 3DP rather than dedicated strategies? What are the characteristics of those components?
- (ii) How much 3DP capacity should a firm procure? Can even marginal investments offer value?
- (iii) What types of supply chains are most likely to benefit from 3DP? Which are least likely?

Answering these questions requires modeling that explicitly accounts for capacity limits and the correlation across component demands and disruptions.

Traditionally, 3DP has *not* been seen as a viable resilience strategy. It often incurs high per-unit production costs, slower production speeds, and expensive capital investments. Research in operations management has largely focused on using 3DP's in low-volume manufacturing, such as spare parts inventory (Song and Zhang 2020, Zhang et al. 2022, Westerweel et al. 2021) and product customization (Chen et al. 2021, Sethuraman et al. 2023).

Outside academia, however, some firms have begun exploring 3DP as a supply chain resilience measure. During the COVID-19 pandemic, CNH Industrial—a major farm equipment manufacturer—experienced a disruption of a small but critical clamping fixture. Halting assembly until replacement parts arrived would have incurred an estimated loss of €189,000 (Materialise 2023). Instead, CNH printed the missing fixture. Although per-unit printing costs were more than seven times higher than injection molding, total printing costs were just €806, the parts were available within a week, and production delays were minimal (Materialise 2023). This suggests that despite its criticality, the fixture had not been backed up in advance—likely because dedicated protection was not economically justified. Following this experience, CNH has institutionalized this 3DP-based resilience approach by developing a “digital warehouse” of printable parts, extending 3DP from one-off emergency fixes to a broader backup capability against supply disruptions (Davies 2023).

A similar pattern arose at General Motors. In September 2021, a design change to the close-out seal of the spoiler for the 2022 Chevy Tahoe necessitated 60,000 new parts within six weeks. Producing the part via injection molding required twelve weeks (McEachern 2022). GM instead printed the required components, meeting the production timeline despite higher per-unit costs (Stevens 2022). More broadly, GM has since invested in 3DP facilities to explore similar applications (Davies 2025).

Together, these examples highlight settings in which 3DP substitutes not for primary production, but for costly or infeasible dedicated backup. Similar trends have also been reported across a wide range of industries. As Dave Prezzano, Global Head of Go-to-Market at HP Personalization and 3D Printing, notes, “recent focus on mitigating supply chain volatility has pushed firms across a wide range of industries toward adopting additive manufacturing (3DP) solutions” (Kubiak 2024). Technology providers such as HP report increasing collaborations with manufacturers to deploy 3DP solutions, including initiatives enabling 3DP mass production for automotive manufacturers such as Volkswagen and BMW, as well as supply chain applications developed with consumer goods companies such as L'Oréal (HP Inc. 2018, Kubiak 2021).

These industrial trends are also reflected in substantive policy initiatives and strategic investments at the national level. A prominent example is the White House’s *AM Forward* initiative, which identifies 3DP as a key technology for strengthening supply chain resilience through flexible production and rapid responses to disruptions. The program brings together major OEMs—including GE Aviation, Honeywell, and Lockheed Martin—to accelerate the adoption of 3DP among small and mid-size manufacturers across U.S. supply chains (The White House 2022). The U.S. Department of Defense has likewise identified 3DP as a key tool for strengthening supply chain resilience. In its report *Securing Defense-Critical Supply Chains* (U.S. Department of Defense (2022)), 3DP is highlighted as an advanced manufacturing capability that can help mitigate supply disruptions and enhance the resilience of the defense industrial base. Recent industry analyses estimate that the DoD alone spent roughly \$300 million on 3DP in 2023, with projected spending expected to grow substantially to \$1.8 Billion in the coming decade (Additive Manufacturing Research 2023).

Our Model: To better understand this disconnect, we introduce a model of 3DP-enabled supply chain resilience in which a firm sources multiple components from multiple suppliers to produce multiple final products. The supplier–component–product relationships form an arbitrary tripartite graph (see Fig. 1). Each supplier is subject to random disruptions (modeled via random yield), while demand for final products is stochastic; both demand and disruptions are arbitrarily correlated with each other and across components. Prior to sourcing, the firm chooses a backup strategy for each component—either a dedicated backup or pooled 3DP capacity—to mitigate potential shortfalls.

To capture the key trade-offs between the two strategies, we model dedicated backup as having unlimited capacity and a small per-unit cost of production, but incurring a fixed cost per component it protects. By contrast, 3DP requires only a single capacity investment to protect many products, but the per unit cost of production is higher and the total recoverable demand is limited by this capacity. To facilitate an apples-to-apples comparison, we model both backup strategies as make-to-order.

Our remaining modeling choices intentionally favor dedicated backup over 3DP. For example, we neglect assume dedicated backups can be brought online instantaneously. These choices allow us to conservatively assess when 3DP adds value by isolating the benefits of pooling and flexibility for the strategic design of the backup portfolio.

Our Contributions: Our central managerial message is that fully leveraging 3DP as a backup strategy requires a portfolio-design perspective, not a component-by-component screening perspective. Conventional wisdom encourages managers to prioritize backing up suppliers using “marginal” attributes; for example, Saghafian and Van Oyen (2012) argue that firms should prioritize backup for

suppliers with longer but less frequent disruptions and/or lower demand uncertainty. Such marginal perspective aligns well with dedicated backup strategies because they are decoupled. By contrast, because 3DP backup “pools” demand shortfalls, adding a component to the 3DP set can either diversify the portfolio (increasing the effectiveness of pooling) or congest it (forcing components to compete for capacity in the same disruption scenarios). Effectively utilizing 3DP as a resilience strategy hence requires a shift in perspective and new decision support tools.

To establish these insights, we first analyze the structure of an optimal backup portfolio. The sets of components backed up by 3DP and by dedicated strategies form a partition (Lemma 1.) Moreover, the components backed up by 3DP are a (potentially strict) superset of those that are unprotected in the absence of 3DP. We derive simple, computable upper and lower bounds on first-stage order quantities by comparing to proxy systems with infinite 3DP capacity and with no 3DP capacity, respectively (Theorem 1). Finally, we bound the optimal printing capacity by a quantile of the aggregate demand shortfall in those two proxy systems (Theorem 2). These bounds already highlight why a portfolio perspective is necessary: optimal 3DP capacity depends on aggregate component shortfall, and, thus, their *joint* distribution. Theorem 3 sharpens this insight by showing that comonotonic shortfalls are worst for 3DP, precisely because there is no benefit to pooling demand shortfalls.

We next turn to developing decision-support tools for constructing such backup portfolios. Computing the firm’s optimal policy requires solving a mixed-binary stochastic optimization problem. Although we can describe an exact solution approach using Bender’s cuts (Section 4), unfortunately, for even moderately sized supply chains, this exact approach is computationally prohibitive. The key challenge is that the problem is neither submodular nor supermodular in the set of components backed up by 3DP (Proposition 3).

Consequently, we develop a scalable algorithmic approach that computes high-quality feasible solutions. Motivated by the “a little flexibility is enough” intuition, we approximate expected second-stage costs by a Taylor expansion around zero printing capacity. The resulting approximation is a global lower bound, becomes tight as capacity vanishes, and is supermodular but non-monotone. While non-monotonic supermodular minimization is NP-Hard, it admits highly efficient approximation algorithms (Feige et al. 2011). Our procedure uses one such approximation algorithm to identify promising candidate 3DP portfolios, and then applies projected stochastic gradient descent to evaluate these candidates under the original stochastic objective. In practice, firms could instead evaluate these candidates using a high-fidelity digital twin.

Of course, in lieu of overhauling their existing resilience strategy in favor of the optimal strategy, many firms may prefer to introduce 3DP incrementally by first backing up components currently

unprotected by dedicated resources. In this case, we derive a necessary and sufficient condition under which an arbitrarily small 3DP capacity investment yields positive cost savings, providing concrete guidance for when piloting 3DP at limited scale is merited (Section 6).

Finally, we use our algorithm to translate this portfolio perspective into managerial guidance for deciding when a component backed up by a dedicated strategy should be switched to 3DP. Leveraging the lower bound above, we propose several *portfolio-level* features that are simple to estimate from simulated or historical scenarios *without* solving the full optimization problem (Section 6.2). These features assess whether switching a component diversifies the existing 3DP portfolio or instead congests it by forcing components to compete for scarce 3DP capacity in the same disruption scenarios. They therefore provide practical screening features for identifying promising switching candidates. Section 7.5 empirically validates that these portfolio features are informative for predicting optimal backup-switching decisions. Again, this portfolio viewpoint contrasts with conventional managerial intuition for dedicated backup strategies which evaluate products through marginal attributes only.

Empirical Evidence: We conclude with an extensive numerical case study calibrated using real-world bill-of-lading data, published 3D printing specifications, and web-scraped prices for Mattel, a global leader in toy manufacturing. The results validate and reinforce our theoretical and managerial insights. Our proposed algorithms consistently compute near-optimal solutions, often within a few percent of the exact optimum. Under current cost estimates, 3DP typically backs up 40–50% of components, including all previously unprotected components (approximately 20–40% of the total) and roughly 20% of components previously protected by dedicated strategies. Even small investments in 3DP capacity generate meaningful benefits: for example, a capacity investment equal to just 5% of total demand yields a 4% cost reduction while cutting demand shortfalls by roughly 50%. In particular, the primary value of 3DP lies in mitigating tail shortfalls, with modest capacity investments substantially reducing the likelihood of severe unmet demand.

Summary: We summarize our contributions as follows:

- In Section 2, we propose a model of a firm's resilience strategy choices capturing the essential trade-offs between 3DP and dedicated backup strategies. We derive structural features of the optimal policy, including which products are backed up by 3DP and bounds on both order quantities and the optimal capacity investment.
- In Section 4 we formulate a mixed-binary stochastic optimization problem for firm's optimal policy. Direct solution is computationally prohibitive. We instead propose a heuristic algorithm

leveraging a Taylor series approximations, supermodularity, and a polishing step in Section 5. This heuristic approach provides high-quality solutions and scales well.

- In Section 6, we develop managerial guidance for incrementally integrating 3DP into an existing resilience portfolio. We derive a necessary and sufficient condition under which even an arbitrarily small 3DP capacity investment yields cost savings, and derive our portfolio features that help identify promising candidates components to switch from dedicated backup to 3DP. These features effectively measure whether a component diversifies or congests the existing 3DP portfolio.
- Finally, in Section 7, we evaluate our algorithm and the value of 3DP through experiments calibrated to Mattel data. Our algorithm yields solutions within a few percent of optimal for moderate chains; for Mattel's full chain, even small 3DP capacity investments yield substantive cost savings and reduce demand shortfalls. Portfolio-level features also materially improve prediction of backup-switching decisions, validating the portfolio-design viewpoint developed in Section 6.

1.1. Additional Related Literature

Much of the existing OM literature on 3DP focuses on low-volume manufacturing applications, such as spare parts (Song and Zhang 2020, Zhang et al. 2022, Westerweel et al. 2021) and product customization (Chen et al. 2021, Sethuraman et al. 2023), likely because these settings require only modest 3DP capital investment. By contrast, we study 3DP as a resilience strategy in supply chains, with an emphasis on identifying which products are best suited for 3DP backup and which supply chains benefit most from pooled flexibility, even if necessitating larger capital investment.

In context of supply chain, Arbabian and Wagner (2020) consider co-locating printers closer to demand. Co-location avoids transportation costs and upstream holding costs, and makes the system more responsive to demand. They quantify these benefits (and how they depend on problem parameters) in a one manufacturer/one retailer setting with a single product. In our paper, we explicitly neglect the benefits of co-location and treat a chain with multiple suppliers to focus exclusively on the benefits of flexibility from 3DP.

Dong et al. (2022) also study 3DP flexibility, specifically how introducing 3DP affects the optimal product offering in design-intensive industries (home furnishing, apparel, jewelry). They focus on three aspects of 3DP printing: i) Design Freedom – 3DP can create products not possible with traditional techniques ii) Quality – 3DP may produce a higher or lower quality product than traditional techniques iii) Flexibility – 3DP can print multiple different types of products in a single run. Those authors develops structural insights on the optimal product offering under a particular choice model for demand and how the above features affect that assortment. Our work adds to this study of 3DP

flexibility by considering a large, disaggregated supply chain. Motivated by the previous anecdotes, our primary focus is on low-cost, low-margin components – not design-intensive products. Hence, issues around customer perceptions of quality and design freedom are arguably second order relative to quantifying the needed capital investment in 3DP and identifying the “right” products to backup.

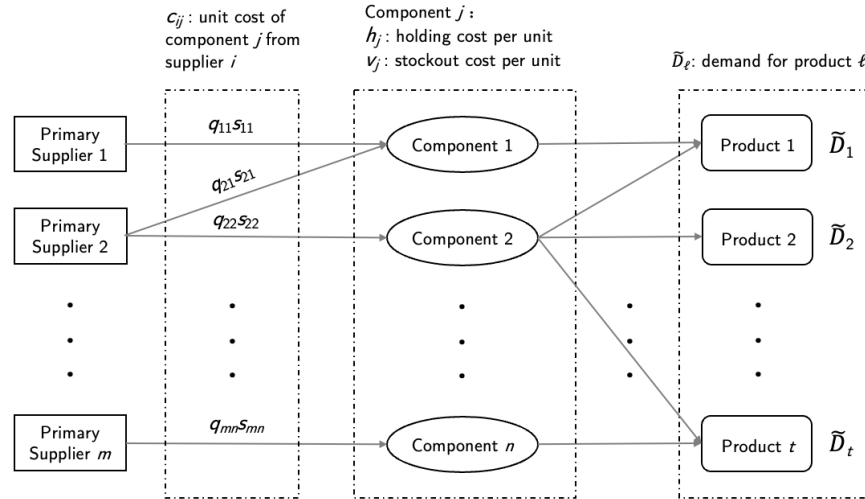
We also connect to the broader work studying flexible backup strategies in supply chain including Saghafian and Van Oyen (2016, 2012). Saghafian and Van Oyen (2016) study a multi-product, multi-supplier network with a rich, Markov Chain model of supply chain disruption that can capture heterogeneous rates of disruption and lengths of disruption. Their focus is on establishing that “a little flexibility” is sufficient to capture the benefits of a fully-flexible backup system. The model studied, however, is very general. While it can in principle be solved via an infinite dimensional Bellman equation, in practice, solving such problems is notoriously challenging. The authors do not offer a specialized algorithm for this task. Consequently, for the most part, the work does not study the question of *which* suppliers to back up with the flexible resource, which is a key question in our work. Indeed, the only results around choosing suppliers for flexible backup from Saghafian and Van Oyen (2016) are with respect to a simplified model with only 2 suppliers. Our work on the other hand adopts an admittedly coarser model of disruptions (random yield) but seeks to develop a general purpose algorithm for identifying which suppliers to back up with 3DP. We demonstrate empirically that our algorithm scales gracefully to very large supply chains.

Finally, we contrast our work to Wang and Webster (2022). In that work, authors also study backup flexibility, but consider a flexible backup that might fail. By considering a 2 product system, they show that flexibility is not always valuable and one must consider whether to invest in flexibility amongst primary suppliers, or amongst backups. The notion of flexibility in this work is generic, meant to represent “supplier development efforts.” In our work, we focus explicitly on 3DP, and, consequently, model it as a *reliable* but expensive backup strategy. We again focus on a multiproduct system where the challenge is identifying which products to backup.

2. Model Setup

We consider a firm that sources n components (indexed by $j \in \mathcal{N} \equiv \{1, \dots, n\}$) from m suppliers to manufacture t final products. The system is represented as a tripartite network linking suppliers, components, and products (Fig. 1). Suppliers may provide multiple components; components may be sourced from multiple suppliers; products may require multiple components; and a given component may be used in multiple products. In other words, the network structure is arbitrary. Without loss of generality, we assume that each product consumes one unit of each required component.

Figure 1 Tripartite representation of the manufacturing network without resilience strategies

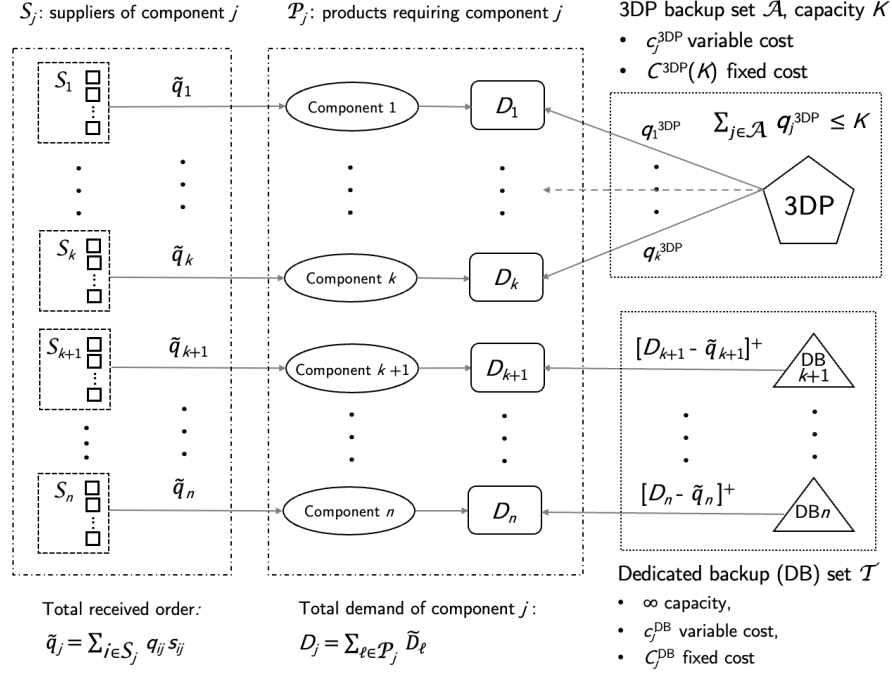


Without Disruptions: In the absence of supply chain disruptions, the firm orders q_{ij} units of component j from each primary supplier $i \in \mathcal{S}_j$ at unit cost c_{ij} , where $\mathcal{S}_j$ denotes the set of suppliers for j . After observing stochastic product demand $\{\tilde{D}_\ell : \ell = 1, \dots, t\}$, the firm allocates the received components to fulfill demand, incurring a holding cost h_j per unit of excess inventory and a shortage cost v_j per unit of shortfall for each component j .

Here and throughout, we interpret the shortfall cost v_j as the cost of extraordinary recovery actions—such as expediting, drop-shipping, emergency fabrication, or manual intervention—undertaken after demand and supply uncertainty is realized. This interpretation reflects settings in which, once the initial orders $\{q_{ij}\}_{i \in \mathcal{S}_j}$ are placed, the firm is effectively committed to producing $\sum_{i \in \mathcal{S}_j} q_{ij}$ units of component j , resorting to extraordinary means to cover any shortfall if disruptions occur. Such recovery actions are naturally defined at the component level and are common in practice, including the emergency manufacturing responses described in the CNH and GM case examples discussed earlier.

This modeling choice also simplifies the firm's post-disruption decision-making. Because extraordinary recovery actions are applied at a uniform per-unit cost v_j , the firm does not need to decide how to allocate limited component supply across downstream products after a disruption. Any feasible allocation yields the same total recovery cost, and all products can ultimately be produced using extraordinary means if necessary. This reduction allows us to zero-in on how pooled, flexible backup capacity reduces the incidence and magnitude of these extreme recovery costs.

With Disruptions: In this paper, however, we consider a setting with disruptions. Namely, let $s_{ij} \in [0, 1]$ be a random variable representing the yield of supplier i for component j . After ordering, the

Figure 2 A resilience strategy combining dedicated and 3DP backups.

Note. For illustration, here 3DP covers the first k components (i.e., $\mathcal{A} = [k]$), while the rest use dedicated backup.

firm receives $q_{ij}s_{ij}$ units, and only pays for received units. To hedge against this yield uncertainty and the possibility of unmet demand, the firm can choose to invest in one of two resilience strategies for each component j :

- (1) **DEDICATED BACKUPS (DB):** The firm can order q_j^{DB} units from an dedicated backup supplier at a per-unit cost c_j^{DB} . Implementing this strategy also incurs a one-time, upfront fixed cost C_j^{DB} , irrespective of the order quantity. This fixed cost reflects the expense of reserving the backup supplier's capacity in advance to ensure access when needed.
- (2) **3D PRINTING (3DP):** The firm can choose to use a 3D-printer to produce q_j^{3DP} units of component j at a per-unit cost of c_j^{3DP} . Unlike dedicated backups, 3DP has a finite capacity K across all products, i.e., we must have $\sum_{j \in \mathcal{N}} q_j^{3DP} \leq K$. We model the cost of investing in 3DP capacity by a non-decreasing, convex function $C^{3DP}(K)$, with $C^{3DP}(0) = 0$, paid before demand and yield are realized. We assume convexity of C^{3DP} to capture diseconomies of scale as higher in-house printing capacity requires additional space, labor, and maintenance. We interpret $C^{3DP}(K)$ as the dominant shared capacity investment; any component-specific digital preparation or qualification costs are typically small relative to this investment (Dong et al. 2022), and hence ignored.

The above choice must be made for each component j . Let $\mathcal{A} \subseteq \mathcal{N}$ denote the set of components backed up using the 3DP strategy, let $\mathcal{T} \subseteq \mathcal{N}$ represent those backed up by DB, and note that a priori these sets need not be disjoint. We seek to choose $\mathcal{A}$ and $\mathcal{T}$ to minimize the total expected costs.

To avoid several trivial scenarios, we will assume throughout that:

ASSUMPTION 1 (Non-trivial Parameters). For each $j \in \mathcal{N}$ and $i \in \mathcal{S}_j$, $c_{ij} < c_j^{\text{DB}} < c_j^{\text{3DP}} < v_j$.

In particular, $v_j \geq \max(c_j^{\text{DB}}, c_j^{\text{3DP}})$ ensures that meeting unmet demand is always preferable to stocking out, while $c_{ij} < \min(c_j^{\text{DB}}, c_j^{\text{3DP}})$ ensures that sourcing from primary suppliers is preferable to expediting or printing. Finally, $c_j^{\text{DB}} < c_j^{\text{3DP}}$ reflects the typical cost relationship between DB and 3DP given current technology.

In summary, the sequence of events are:

- i) The firm chooses sets $\mathcal{A}$ and $\mathcal{T}$ of components backed up by 3DP and DB, respectively, and a capacity $K \in [0, \infty)$ for the 3D printing resource. These choices induce a total fixed cost of $\sum_{j \in \mathcal{T}} C_j^{\text{DB}} + C^{\text{3DP}}(K)$.
- ii) The firm orders q_{ij} units from supplier $i \in \mathcal{S}_j$ for each component $j \in \mathcal{N}$.
- iii) The firm observes random yields s_{ij} and demands D_j (aggregates demand $\tilde{D}_\ell$ for all products $\ell \in \mathcal{P}_j$ requiring j). It pays $\sum_{j \in \mathcal{N}} \sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} s_{ij}$ for all successful deliveries.
- iv) The firm orders q_j^{DB} units from the expedited supplier for each $j \in \mathcal{T}$ and prints q_j^{3DP} units on the 3D printer, subject to $\sum_{j \in \mathcal{A}} q_j^{\text{3DP}} \leq K$, inducing a cost of $\sum_{j \in \mathcal{T}} c_j^{\text{DB}} q_j^{\text{DB}} + \sum_{j \in \mathcal{A}} c_j^{\text{3DP}} q_j^{\text{3DP}}$.
- v) The firm pays extraordinary means costs v_j per unit of any shortfall and holding costs h_j per unit of any excess production for each component $j \in \mathcal{N}$.

Our goal is to minimize the total expected costs, i.e.,

$$\min_{\mathcal{T}, \mathcal{A} \subseteq \mathcal{N}, K \geq 0} C^{\text{3DP}}(K) + \sum_{j \in \mathcal{T}} C_j^{\text{DB}} + U^{\text{3DP}}(\mathcal{A}, K) + U^{\text{DB}}(\mathcal{T}) \quad (2.1)$$

where $U^{\text{DB}}(\mathcal{T})$ and $U^{\text{3DP}}(\mathcal{A}, K)$ represent the expected operational costs incurred from iii) to v). These terms are described by optimization problems over $q_{ij}, q_j^{\text{3DP}}, q_j^{\text{DB}}$. For completeness, we state them now, but provide detailed derivations and commentary on these problems in Sections 2.1 and 2.2 below:

$$U^{\text{DB}}(\mathcal{T}) \equiv \sum_{j \in \mathcal{T}} U_j^{\text{DB}},$$

$$U_j^{\text{DB}} \equiv \min_{q_j \geq 0} \mathbb{E}_{D_j, s_j} \left[\sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} s_{ij} + c_j^{\text{DB}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^+ + h_j \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^- \right], \quad (2.2)$$

$$U^{3DP}(\mathcal{A}, K) \equiv \min_{\mathbf{q} \geq 0} \sum_{j \in \mathcal{A}} \sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} \mathbb{E}[s_{ij}] + \mathbb{E}_{\mathbf{D}, \mathbf{s}} [V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, \mathcal{A}, K)], \quad (2.3)$$

where

$$\mathbf{q} \circ \mathbf{s} \equiv \left(\sum_{i \in \mathcal{S}_1} q_{i1} s_{i1}, \sum_{i \in \mathcal{S}_2} q_{i2} s_{i2}, \dots, \sum_{i \in \mathcal{S}_n} q_{in} s_{in} \right)^\top \in \mathbb{R}^n, \quad \text{and,}$$

$$V^{3DP}(\mathbf{y}, \mathcal{A}, K) \equiv \min_{\mathbf{q}^{3DP} \geq 0} \sum_{j \in \mathcal{A}} \left(c_j^{3DP} q_j^{3DP} + v_j [y_j - q_j^{3DP}]^+ + h_j [y_j - q_j^{3DP}]^- \right)$$

$$\text{s.t. } \sum_{j \in \mathcal{A}} q_j^{3DP} \leq K \quad (2.4)$$

Before proceeding, we prove three simple properties of the optimal solution that we use in these derivations and throughout: First, because $c_j^{\text{DB}} < v_j$ and the expedited backup is unconstrained, it is always optimal to expedite all units of unmet demand for $j \in \mathcal{T}$, i.e., $q_j^{\text{DB}} = [D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$. Second, since $c_j^{\text{DB}} < c_j^{3DP}$ and expedited backup is unconstrained, we always prefer to expedite unmet demand from a supplier $j \in \mathcal{T}$ over printing it. Hence, $\mathcal{A} \cap \mathcal{T} = \emptyset$. Finally, since we are always free to choose $q_j^{3DP} = 0$ for any $j \in \mathcal{A}$, we can without loss of generality take $\mathcal{A} = \mathcal{T}^c$. We summarize these last two observations in the following proposition:

LEMMA 1 (Backup Strategies Form a Partition). *Let $\mathcal{T}^*$ and $\mathcal{A}^*$ be optimal for (2.1). Then, $\mathcal{T}^* = (\mathcal{A}^*)^c$.*

Thus, in the remainder, we let $\mathcal{T} = \mathcal{A}^c$ and work with the set $\mathcal{A}$. Furthermore, when $\mathcal{A}$ and K are fixed and clear from context, we omit them from notation for brevity.

2.1. Dedicated Backups and $U^{\text{DB}}(\mathcal{A}^c)$

We next present the derivation of U^{DB} in Problem (2.2). By construction, dedicated backups decouple across products, i.e., $U^{\text{DB}}(\mathcal{A}^c) = \sum_{j \in \mathcal{A}^c} U_j^{\text{DB}}$, where U_j^{DB} includes primary ordering, expediting, and holding costs. Recall, under Assumption 1, any unmet demand $[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$ is fully covered by DB's unlimited capacity and there are no stock-outs. This gives rise to the newsvendor-type problem in Problem (2.2), which can be addressed using standard techniques from stochastic convex optimization, such as stochastic first-order methods. When component j has a unique supplier i , the optimization problem admits an explicit solution, as stated in the lemma below. The characterization relies on the following quantity:

$$\bar{q}_{ij}(z, D) \equiv \inf \left\{ q \geq 0 : \mathbb{E}_{D, s_{ij}} [s_{ij} \mathbb{I}\{D \leq q s_{ij}\}] \geq \left(\frac{z - c_{ij}}{z + h_j} \right) \mathbb{E}[s_{ij}] \right\}. \quad (2.5)$$

Here, with a slight abuse of notation, D in $\bar{q}_{ij}(z, D)$ denotes the probability distribution of demand rather than a particular realization, so that $\bar{q}_{ij}$ is a deterministic quantity.

LEMMA 2 (First-Stage Ordering for DB). *Let i be the unique supplier of component j , then the optimal solution of Problem (2.2) with the smallest magnitude is given by $\bar{q}_{ij}(c_j^{\text{DB}}, D_j)$, where:*

In special cases, $\bar{q}_{ij}$ reduces to the newsvendor quantile, e.g., if D_j and s_{ij} are independent and s_{ij} is binary. Otherwise, one must solve Eq. (2.5) numerically, e.g., by bisection search on q . Later, we will use $\bar{q}_{ij}(\cdot, \cdot)$ to construct bounds on the optimal first-stage order for j backed up by 3DP.

2.2. 3DP and $U^{\text{3DP}}(\mathcal{A}, K)$

We now present the derivation of U^{3DP} , c.f. Problem (2.3). Unlike dedicated strategies, the optimal 3DP backup quantities $\mathbf{q}^{\text{3DP}}$ lack a simple, closed-form due to the capacity constraint. Instead, U^{3DP} is defined by a two-stage stochastic optimization problem. The function V^{3DP} in Problem (2.4) represents the second-stage recourse cost that minimizes printing, holding, and stock-out costs with respect to the printing quantities $\mathbf{q}^{\text{3DP}}$ after the realization of $\mathbf{q}$, $\mathbf{s}$, and $\mathbf{D}$. Using standard results from the stochastic programming literature, we can show that computing q_j for $j \in \mathcal{A}$ amounts to solving a convex optimization problem:

LEMMA 3 (Convexity of U^{3DP} and V^{3DP}). *The following holds for U^{3DP} and V^{3DP} :*

- (i) *The function $V^{\text{3DP}}(\mathbf{D} - \mathbf{q} \circ \mathbf{s})$ is convex in $\mathbf{q}$ for any given $(\mathbf{s}, \mathbf{D})$*
- (ii) *Problem (2.3) (which defines U^{3DP}) is convex.*
- (iii) *The function $K \mapsto U^{\text{3DP}}(\mathcal{A}, K)$ is convex in K for any $\mathcal{A}$.*

While many algorithms exist for solving convex problems, later in Section 5.4, we present a simple approach that leverages the specific structure of $V^{\text{3DP}}(\cdot)$ to efficiently solve Problem (2.3) and integrate it into a scalable heuristic framework for optimizing the first-stage decisions $\mathcal{A}$ and K .

2.3. Additional Model Discussion

We view the 3D printer as a flexible, make-to-order manufacturing resource capable of producing any component. In reality, some component, however, may not be printable (using current technology) because of their engineering specifications. We exclude them from $\mathcal{N}$ because they do not affect the choice of $\mathcal{A}$, and computing their optimal order quantities and dedicated backup quantities can be done using simple newsvendor-like calculations, outside of the model, see Section 2.1.

Note that we also make no assumptions on the exogenous dependence structure of product demands $\tilde{D}_\ell$ or supplier yields s_{ij} . That said, the general manufacturing network we consider (cf. Fig. 1 and Fig. 2) naturally induces dependencies at the component level. Specifically, the realized supply of component j , given by $\sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}$, may be correlated with that of any component j' that

shares common suppliers with j , as the yields s_{ij} from the same supplier i are typically highly correlated. Similarly, the aggregated demand $D_j = \sum_{\ell \in \mathcal{P}_j} \tilde{D}_\ell$ may be correlated with the demand for any component j' that appears in overlapping products with j .

The lead-time advantage of 3DP is captured implicitly through the timing of recourse decisions and the magnitude of costs. Specifically, our model is best viewed as a three-stage model: the firm first designs its backup portfolio, then places primary orders, and finally chooses printing and other recovery actions after demand and disruptions are realized. In particular, 3DP production is chosen after disruptions, but delivered in time to meet demand, reflecting its role as a rapid-response technology. Further, v_j reflects not only the direct cost of last-minute recovery actions—such as emergency expediting or ad hoc fabrication—but also delay-related costs incurred when no advance preparation or fast-response capability is available. Admittedly, this is a simplified model of downstream recourse decisions and costs, but it captures the first-order tradeoffs relevant to our primary research question: how 3DP changes the ex ante design of a portfolio of backup strategies.

Finally, while we have described dedicated backup as standby suppliers secured in advance for ease of exposition, it can easily represent any dedicated strategy that entails a fixed cost and a cheaper per-unit cost, including employing dual sites or purchasing buffering inventory.

3. Properties of Optimal 3DP Strategies

In this section we present properties of the optimal backup strategy by analyzing the stochastic optimization problem Problem (2.1). Many of structural insights apply to any candidate 3DP backup set, including suboptimal sets dictated by exogenous managerial considerations.

We first define the set $\mathcal{A}_0$ of unprotected components in the absence of 3DP, i.e., components that did not merit a dedicated backup. Unlike the optimal 3DP backup set $\mathcal{A}^*$ which may be difficult to compute, $\mathcal{A}_0$ is easily identified: Because dedicated backups decouple across components, a component $j \in \mathcal{N}$ is in $\mathcal{A}_0$ if and only if its costs when protected $U_j^{\text{DB}} + C_j^{\text{DB}}$ are larger than its costs when unprotected. These unprotected costs are given by solving Problem (2.2) after replacing c_j^{DB} by v_j , because there are no associated fixed costs for an unprotected product.

The next proposition relates $\mathcal{A}_0$ with the optimal 3DP backup set $\mathcal{A}^*$:

PROPOSITION 1 (Unprotected Products and Optimal Backup). $\mathcal{A}_0 \subseteq \mathcal{A}^*$ and this containment can be strict.

Thus, introducing 3DP backup impacts a firm's resilience practices in two ways: it covers all previously unprotected components *and* it *may* switch some components from dedicated to 3DP backup.

Proposition 1 has important implications for a firm piloting the use of 3DP as a backup strategy. Since $\mathcal{A}_0$ is easy to compute, such a firm might consider only backing up $\mathcal{A}_0$ instead of computing $\mathcal{A}^*$. While suboptimal, this strategy avoids the need to renegotiate terms with any existing expediting suppliers used by dedicated backup while the firm builds internal expertise around 3DP, and functions as a natural stepping stone towards backing up the entire set $\mathcal{A}^*$ later. In Section 6, we further explore the perspective of this firm piloting 3DP as a backup strategy and provide necessary and sufficient conditions for this suboptimal strategy to be profitable with even a small capacity investment. In the remainder of this section, we develop properties of the optimal solution for a generic $\mathcal{A}$, since we might be interested in $\mathcal{A}_0$ or $\mathcal{A}^*$, or some other set between these two.

For any choice of 3DP backups $\mathcal{A}$ and 3DP capacity K , we must identify corresponding first-stage order quantities $q_{ij}^* = q_{ij}^*(\mathcal{A}, K)$ for all $i \in \mathcal{S}_j$ and $j \in \mathcal{A}$. Although computing these quantities exactly requires solving a two-stage convex optimization problem, Theorem 1 provides bounds that can easily be computed via simple bisection search.

THEOREM 1 (Bounds on Optimal First-Stage Orders). *For any set $\mathcal{A}$ and $K \geq 0$, let $\mathbf{q}^*(\mathcal{A}, K)$ be the optimal solution to $U^{\text{3DP}}(\mathcal{A}, K)$ (c.f. Problem (2.3)) with minimal ℓ_1 -norm.¹ Let*

$$\bar{q}_{ij}^0 \equiv \bar{q}_{ij}(v_j, D_j) \quad \text{and} \quad \bar{q}_{ij}^\infty \equiv \bar{q}_{ij}\left(c_j^{\text{3DP}}, D_j - \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \bar{q}_{\ell j}^0 s_{\ell j}\right)$$

where $\bar{q}_{ij}(\cdot, \cdot)$ is defined in Eq. (2.5). Then, for all $i \in \mathcal{S}_j$ and $j \in \mathcal{A}$, we have $\bar{q}_{ij}^\infty \leq q_{ij}^*(\mathcal{A}, K) \leq \bar{q}_{ij}^0$.

In the same spirit as Lemma 2, $\bar{q}_{ij}^0$ denotes the smallest optimal first-stage order of component j from supplier $i \in \mathcal{S}_j$ when it is the sole supplier and no backup is available; whereas $\bar{q}_{ij}^\infty$ denotes the smallest optimal order when we order $\bar{q}_{\ell j}^0$ from the other suppliers $\ell \in \mathcal{S}_j \setminus \{i\}$ and component j is under 3DP protection with infinite capacity $K = \infty$. We stress, the bounds hold for each $i \in \mathcal{S}_j$ and $j \in \mathcal{A}$. (We will use this property later when designing our algorithms.) Finally, we note that while computing $q_{ij}^*(\mathcal{A}, K)$ requires knowledge of the full joint-distribution of $(D_j, \mathbf{s}_j)$ across components $(\mathbf{s}_j \equiv (s_{ij} : i \in \mathcal{S}_j))$, computing the bounds above only requires the marginal distribution of each $(D_j, \mathbf{s}_j)$ pair. Calibrating such a distribution to data may be substantially easier in practice.

Our above bounds can also be used to estimate the optimal 3DP capacity investment. Specifically, let $K^*(\mathcal{A})$ be the minimizer of Problem (2.1) for a given $\mathcal{A}$ and $K^*(\mathcal{A}, \mathbf{q})$ be the optimal capacity for a given $\mathcal{A}$ and fixed first-stage order $\mathbf{q} \geq 0$. In both cases, if the corresponding problem admits multiple optima, we define $K^*(\mathcal{A})$ or $K^*(\mathcal{A}, \mathbf{q})$ as the minimum optimal solution.

¹ This choice is simply to tie-break among multiple optima.

THEOREM 2 (Bounds on Optimal 3DP Capacity). *Let $C^{3DP}(K) = c^{\text{cap}} K$ for some $c^{\text{cap}} > 0$.*

- (i) *Suppose there exists an $r > 0$ such that $v_j - c_j^{3DP} = r$ for all $j \in \mathcal{A}$, then $K^*(\mathcal{A}, \mathbf{q})$ is the $(1 - \frac{c^{\text{cap}}}{r})$ -quantile of $\sum_{j \in \mathcal{A}} [D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$, for any $\mathbf{q} \geq 0$.*
- (ii) *Let $r^{\min} \equiv \min_{j \in \mathcal{A}} (v_j - c_j^{3DP})$ and $r^{\max} \equiv \max_{j \in \mathcal{A}} (v_j - c_j^{3DP})$. Then, $K^*(\mathcal{A})$ is bounded below by the $(1 - \frac{c^{\text{cap}}}{r^{\min}})$ -quantile of $\sum_{j \in \mathcal{A}} [D_j - \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^0 s_{ij}]^+$ and bounded above by the $(1 - \frac{c^{\text{cap}}}{r^{\max}})$ -quantile of $\sum_{j \in \mathcal{A}} [D_j - \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^\infty s_{ij}]^+$.*

Part (i) of the theorem characterizes the optimal capacity $K^*(\mathcal{A}, \mathbf{q})$ as a quantile of the total demand shortfall $\sum_{j \in \mathcal{A}} [D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$. It therefore depends on the joint distribution of $(D_j, \mathbf{s}_j)$ across *all* components in $\mathcal{A}$. Part (ii) extends this characterization to heterogeneous margins by combining the ordering bounds from Theorem 1 with extremal values of $v_j - c_j^{3DP}$ to obtain computable upper and lower bounds on $K^*(\mathcal{A})$.

Theorem 2 also highlights how cross-component dependence shapes the value of 3DP. When demand shortfalls are weakly dependent, the relevant quantiles of the total shortfall tend to be small, and only modest 3DP capacity is required. In contrast, strong positive dependence can substantially increase these quantiles, making large capacity investments necessary and potentially unattractive. We can formalize this last statement using the supermodular ordering of random variables. Recall that random variables $X_1, \dots, X_n$ are comonotonic if there exist nondecreasing functions f_i , $i = 1, \dots, n$, and a common random variable U such that $(X_1, \dots, X_n) \sim_d (f_1(U), \dots, f_n(U))$.

THEOREM 3 (Comonotonic Shortfalls Are the Worst). *Suppose the marginal distributions of $(\mathbf{D}, \mathbf{s})$ are fixed. Then, the optimal cost of Problem (2.1) is maximal when the joint distribution of $(\mathbf{D}, \mathbf{s})$ is such that the shortfalls $[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$ are comonotonic across $j \in \mathcal{N}$ for all $\mathbf{q} \geq 0$.*

In words, 3DP-based resilience is least effective when shortfalls are as positively dependent as possible, i.e., they are comonotonic.

Theorem 3 highlights a key distinction between 3DP and dedicated resilience strategies. Because dedicated backups decouple across components by construction, their cost depends only on the marginal distributions of each $(D_j, \mathbf{s}_j)$ pair and is invariant to cross-component dependence. In contrast, the effectiveness of 3DP hinges critically on joint dependence. This shifts the resilience problem from evaluating components in isolation to managing a portfolio of risks, and implies that effective 3DP backup sets $\mathcal{A}$ should favor components whose shortfalls are weakly dependent or anti-correlated, rather than tightly coupled. We explore this idea more fully in Section 6.

4. Optimizing the 3DP Strategy via Mixed-Integer Optimization

In this section, we compute the optimal $\mathcal{A}^*$, $\mathbf{q}^*$ and K^* exactly, instead of bounding them, by reformulating (2.1) as a mixed-integer optimization (MIO) problem.

4.1. Mixed Integer Optimization Reformulation

For any $\mathcal{A} \subseteq \mathcal{N}$, we define a corresponding binary vector $\mathbf{x} \in \{0, 1\}^n$ such that $x_j = 1$ if $j \in \mathcal{A}$ and $x_j = 0$ otherwise. We can then rewrite U^{3DP} and V^{3DP} as functions in $\mathbf{x}$:

$$U^{3DP}(\mathbf{x}, K) \equiv \min_{\mathbf{q} \geq 0} \sum_{j \in \mathcal{N}} \sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} x_j \mathbb{E}[s_{ij}] + \mathbb{E}_{\mathbf{D}, \mathbf{s}} [V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, \mathbf{x}, K)], \quad (4.1)$$

$$V^{3DP}(\mathbf{y}, \mathbf{x}, K) \equiv \min_{\mathbf{q}^{3DP} \geq 0} \sum_{j \in \mathcal{N}} \left(c_j^{3DP} q_j^{3DP} + v_j [y_j - q_j^{3DP}]^- + h_j [y_j - q_j^{3DP}]^+ \right) x_j \quad (4.2)$$

$$\text{s.t. } \sum_{j \in \mathcal{N}} q_j^{3DP} x_j \leq K.$$

This reformulation, although straightforward, introduces bilinear terms like $q_{ij}x_j$ in the objective, which are notoriously difficult to handle. We prefer a “big M” type formulation in what follows, and develop bounds to ensure the “M” is not too large.

LEMMA 4 (“Big M” Formulation of V^{3DP}). Assume $0 \leq D_j \leq \bar{D}_j$ and $s_{ij}^{\min} \leq s_{ij} \leq s_{ij}^{\max}$ almost surely for all $i \in \mathcal{S}_j$ and $j \in \mathcal{N}$. Then for all $\mathbf{x} \in \{0, 1\}^n$, $K \geq 0$, and $\mathbf{y} = \mathbf{D} - \mathbf{q} \circ \mathbf{s}$:

$$V^{3DP}(\mathbf{y}, \mathbf{x}, K) = \min_{\mathbf{q}^{3DP}, \mathbf{z}^{3DP}} \sum_{j=1}^n \left(c_j^{3DP} q_j^{3DP} + v_j [y_j - z_j^{3DP}]^- + h_j [y_j - z_j^{3DP}]^+ \right) \quad (4.3a)$$

$$\text{s.t. } 0 \leq q_j^{3DP} \leq M_j^1 x_j, \quad \forall j = 1 \dots n, \quad (4.3b)$$

$$-M_j^2(1 - x_j) \leq z_j^{3DP} - q_j^{3DP} \leq M_j^2(1 - x_j), \quad \forall j = 1 \dots n, \quad (4.3c)$$

$$-M_j^3 x_j \leq y_j - z_j^{3DP} \leq M_j^3 x_j, \quad \forall j = 1 \dots n, \quad (4.3d)$$

$$\sum_{j=1}^n q_j^{3DP} \leq K, \quad (4.3e)$$

where for all $j \in \mathcal{N}$, we let

$$M_j^1 = \bar{D}_j - \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^{\infty} s_{ij}^{\min}, \quad M_j^2 = M_j^3 = \max \left\{ \bar{D}_j - \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^{\infty} s_{ij}^{\min}, \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^0 s_{ij}^{\max} \right\}. \quad (4.4)$$

Additionally, V^{3DP} is convex jointly in $(\mathbf{y}, \mathbf{x}, K)$.

It is well-known that the effectiveness of “big M” formulations like Problem (4.3) hinges on the choice of “M”, which in our case correspond to M_j^1 , M_j^2 , and M_j^3 . We have used the bounds from Theorem 1 to choose these values.

We can now use this reformulation of $V^{3DP}(\cdot)$ to reformulate Problem (2.1):

THEOREM 4 (Big “M” formulation for Optimizing $\mathcal{A}$). *Let the assumptions in Lemma 4 holds and let V^{3DP} be as reformulated in Lemma 4. Then, we can determine the optimal backup strategy $\mathcal{A}$, first-stage ordering quantities $\mathbf{q}$, and 3DP capacity K by solving the two-stage, stochastic MIO:*

$$\begin{aligned} \min_{\mathbf{x} \in \{0,1\}^n, K \geq 0, \mathbf{q}} \quad & C^{3DP}(K) + \sum_{j=1}^n (1 - x_j) U_j^{DB} + \sum_{j=1}^n \sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} \mathbb{E}[s_{ij}] + \mathbb{E}_{\mathbf{D}, \mathbf{s}} [V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, \mathbf{x}, K)] \\ \text{s.t.} \quad & \bar{q}_{ij}^\infty x_j \leq q_{ij} \leq \bar{q}_{ij}^0 x_j, \quad \forall j = 1 \dots n. \end{aligned} \quad (4.5)$$

Moreover, the objective function of Problem (4.5) is jointly convex in $\mathbf{x} \in \mathbb{R}^n$, $K \geq 0$ and $\mathbf{q} \geq 0$.

4.2. Solving Problem (4.5) at Scale

In principle, Problem (4.5) can be solved using an off-the-shelf solver (e.g., Gurobi) after approximating the expectation by S scenarios. However, this becomes impractical for large supply chains as the number of variables scales with nS , where n is the number of suppliers. While many variables are expected to be zero when $|\mathcal{A}|$ or K is small, generic branch-and-bound algorithms fail to exploit this sparsity or other structural properties of V^{3DP} .

An alternative approach is Benders decomposition, which replaces the second-stage cost with an auxiliary variable θ and enforces the epigraphic constraint $\mathbb{E} [V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, \mathbf{x}, K)] \leq \theta$, where V^{3DP} is reformulated as in (4.3). The method iteratively tightens this constraint by adding valid cuts. Cut generation requires evaluating a subgradient of $V^{3DP}(\mathbf{y}, \mathbf{x}, K)$, which admits a closed-form expression (cf. Proposition 6). Further details are deferred to Section B.1.

5. Approximations and Supermodularity Heuristics for Computing $\mathcal{A}$

Theorem 4 and Problem (4.5) describe an exact approach for computing the optimal policy $\mathcal{A}^*$. However, even with Benders decomposition, solving the MIP directly can be prohibitively expensive for a large supply chain. We next develop a bound leveraging the intuition that “a little flexibility is enough,” so many firms may invest only modestly in 3DP. The resulting bound is valid for all capacity levels and becomes tight as $K \rightarrow 0$. We illustrate its computational effectiveness in Section 7 through our case-study with Mattel’s supply-chain (c.f. Fig. 5).

Recall, a set function $f : 2^{\mathcal{N}} \mapsto \mathbb{R}$ is supermodular if for all $S \subseteq T$ and $i \notin S$, it holds that:

$$f(T \cup \{i\}) - f(T) \geq f(S \cup \{i\}) - f(S) \quad (5.1)$$

Similarly, $f(\cdot)$ is submodular if $-f$ is supermodular. Intuitively, supermodularity captures increasing return to scale. There is a rich literature on optimizing submodular/supermodular functions. See, e.g., Krause and Golovin (2014). We would ideally leverage these techniques, but, as we next show, our problem is neither submodular nor supermodular.

5.1. Supermodularity of U^{3DP} : Positive and Negative Results

We start with a positive result, the objective function in the optimization defining $U^{3DP}(K, \mathcal{A})$ is supermodular:

PROPOSITION 2. *The function $\mathcal{A} \mapsto \sum_{j \in \mathcal{A}} \sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} \mathbb{E}[s_{ij}] + \mathbb{E}[V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, \mathcal{A}, K)]$ is supermodular in $\mathcal{A} \subseteq \mathcal{N}$ for any fixed $\mathbf{q} \in \mathbb{R}^n$ and $K \geq 0$.*

Proposition 2 suggests that under an alternate simplified model, where first-stage order quantities are specified *exogenously* in a manner that does not depend on $\mathcal{A}$, one enjoys supermodularity. Unfortunately, in our model, first stage costs are chosen endogenously to optimize costs, which necessarily depends on the set $\mathcal{A}$. As a result, $U^{3DP}(K, \mathcal{A})$ is neither supermodular nor submodular.

PROPOSITION 3. *$U^{3DP}(K, \mathcal{A})$ is neither supermodular nor submodular in $\mathcal{A} \subseteq \mathcal{N}$.*

It seems then that optimizing the first stage costs partially drives the computational complexity of our model. This motivates seeking supermodular approximations.

5.2. A Supermodular Approximation of U^{3DP}

We next propose a supermodular approximation of U^{3DP} based on its Taylor expansion in K around $K = 0$. The key intuition is that 3DP essentially serves as “flexibility” in the backup resource, and a host of operations literature suggests “a little flexibility is enough.” Combined with the costliness of printers, we thus intuit an optimal solution should have small K . Collectively, these observations motivate approximating U^{3DP} via its Taylor expansion at small K . The resulting first-order term is both supermodular *and* a lower bound. For convenience, we denote U_j^0 as the optimal value of Problem (2.2) with c_j^{DB} replaced by v_j , i.e., the operational cost for component j without backup, and let $\{\hat{q}_{ij}^0 : i \in \mathcal{S}_j\}$ be an optimal solution to this problem.

THEOREM 5 (Supermodular Lower Bound for $U^{3DP}(K, \mathcal{A})$). *Let*

$$L^{3DP}(\mathcal{A}, K) \equiv \sum_{j \in \mathcal{A}} U_j^0 - K \mathbb{E}_{\mathbf{D}, \mathbf{s}} \left[\max_{j \in \mathcal{A}} (v_j - c_j^{3DP}) \mathbb{I} \left\{ D_j > \sum_{i \in \mathcal{S}_j} \hat{q}_{ij}^0 s_{ij} \right\} \right]. \quad (5.2)$$

Then,

- i) *For any $\mathcal{A}$, $L^{3DP}(\mathcal{A}, K) \leq U^{3DP}(\mathcal{A}, K)$.*
- ii) *This bound becomes tight as K becomes small, i.e., $U^{3DP}(\mathcal{A}, K) - L^{3DP}(\mathcal{A}, K) \rightarrow 0$ as $K \rightarrow 0$.*
- iii) *For any fixed $K \geq 0$, the function $\mathcal{A} \mapsto L^{3DP}(\mathcal{A}, K)$ is supermodular.*

Since $L^{3DP}(\mathcal{A}, K)$ is supermodular in $\mathcal{A}$, a natural heuristic is to minimize this lower bound in lieu of $U^{3DP}(\mathcal{A}, K)$, which should perform well if the optimal K is small. Beyond computation, we

also use this bound in Section 6.2 to approximate the marginal value of switching components to 3DP backup, yielding portfolio-level screening features that inform simple managerial rules and are validated empirically in Section 7.5.

We pursue this heuristic in the next section, but first develop two weaker bounds that may be of interest in specialized settings:

PROPOSITION 4 (Simpler Supermodular Lower Bounds for U^{3DP}). *Let*

$$\widehat{L}^{3DP}(\mathcal{A}, K) \equiv \sum_{j \in \mathcal{A}} U_j^0 - K \left(\max_{j \in \mathcal{N}} v_j - c_j^{3DP} \right) \mathbb{P}(\exists j \in \mathcal{A} \text{ s.t. } D_j > \sum_{i \in \mathcal{S}_j} \widehat{q}_{ij}^0 s_{ij}). \quad (5.3)$$

Then,

- i) $L^{3DP}(\mathcal{A}, K) \geq \widehat{L}^{3DP}(\mathcal{A}, K)$.
- ii) *For any fixed $K \geq 0$, the function $\mathcal{A} \mapsto \widehat{L}^{3DP}(\mathcal{A}, K)$ is supermodular.*
- iii) $\widehat{L}^{3DP}(\mathcal{A}, K) = L^{3DP}(\mathcal{A}, K)$ *whenever the per-unit margin $v_j - c_j^{3DP}$ is identical across $j \in \mathcal{N}$.*

Similarly, for any fixed $\lambda > 0$, define

$$\widetilde{L}^{3DP}(\mathcal{A}, K, \lambda) \equiv \sum_{j \in \mathcal{A}} U_j^0 - \frac{K}{\lambda} \log \left(\sum_{j \in \mathcal{A}} \left[\mathbb{P} \left(D_j \leq \sum_{i \in \mathcal{S}_j} \widehat{q}_{ij}^0 s_{ij} \right) + \mathbb{P} \left(D_j > \sum_{i \in \mathcal{S}_j} \widehat{q}_{ij}^0 s_{ij} \right) e^{\lambda(v_j - c_j^{3DP})} \right] \right). \quad (5.4)$$

Then,

- iv) $L^{3DP}(\mathcal{A}, K) \geq \widetilde{L}^{3DP}(\mathcal{A}, K, \lambda)$ *for all $\lambda > 0$.*
- v) *For any fixed $K \geq 0$, $\lambda \geq 0$, the function $\mathcal{A} \mapsto \widetilde{L}^{3DP}(\mathcal{A}, K, \lambda)$ is supermodular.*

In Eq. (5.4), λ should be interpreted as a hyperparameter that is set exogenously.

In lieu of the expectation, $\widehat{L}^{3DP}(\mathcal{A}, K)$ depends on the probability that there is some unmet demand in $\mathcal{A}$ under the naive ordering strategy $\widehat{q}^0$. Estimating this probability may be easier in practice than estimating the expectation in Eq. (5.2). On the other hand, evaluating this probability still requires the joint distribution of $(D_j, \mathbf{s}_j)$ across j which may be difficult to specify. The looser bound (5.4) avoids this and only depends on the distributions of the pair random variables $(D_j, \mathbf{s}_j)$ for each $j \in \mathcal{A}$. This might be substantially easier to estimate in practice. Thus, depending on data availability, either bound might be preferred to $\widehat{L}^{3DP}$.

As an aside, we note that replacing the maximum over $j \in \mathcal{N}$ in Eq. (5.3) with the maximum over $j \in \mathcal{A}$ also yields a tighter lower bound than $\widehat{L}^{3DP}(\mathcal{A}, K)$, but that bound is no longer supermodular.

Algorithm 1 SuperMod Approx: A Heuristic for Optimizing $\mathcal{A}$ and K

Require: A grid $\mathcal{K}$ of K values

for each fixed $K \in \mathcal{K}$ **do**

 Solve the inner minimization of Problem (5.6) using local search (c.f. Section B.2) for $\hat{\mathcal{A}}(K)$

 Compute $U^{3DP}(\hat{\mathcal{A}}(K), K)$ via Projected Stochastic Gradient Descent.

 Let $F(K)$ be the the objective value of the inner problem in of Problem (5.5).

return $K^* = \arg \min_{K \in \mathcal{K}} F(K)$

5.3. Our Proposed Heuristic for Optimizing $\mathcal{A}$ and K

Our heuristic for optimizing $\mathcal{A}$ and K treats Problem (2.1) as a two-step optimization, and then replaces $U^{3DP}(\mathcal{A}, K)$ by $L^{3DP}(\mathcal{A}, K)$:

$$\min_{K \geq 0} C^{3DP}(K) + \min_{\mathcal{A} \subseteq \mathcal{N}} \left\{ \sum_{j \in \mathcal{A}^c} (C_j^{DB} + U_j^{DB}) + U^{3DP}(\mathcal{A}, K) \right\}, \quad (5.5)$$

$$\implies \min_{K \geq 0} C^{3DP}(K) + \min_{\mathcal{A} \subseteq \mathcal{N}} \left\{ \sum_{j \in \mathcal{A}^c} (C_j^{DB} + U_j^{DB}) + L^{3DP}(\mathcal{A}, K) \right\} \quad (5.6)$$

We solve the outer minimization in Problem (5.6) via grid search over K . For a fixed K , Theorem 5 shows that the inner problem entails minimizing a supermodular function, or, equivalently, maximizing a positive-valued, non monotone submodular function. (The non-monotonicity arises because adding products to the 3DP set may not reduce costs when K is fully utilized.) Feige et al. (2011) proves maximizing such functions is NP-Hard, but also shows that a simple local-search heuristic attains a $\frac{1}{3}$ approximation. Inspired by this result, we also use a local search to optimize the inner problem by iteratively adding or removing a product from $\mathcal{A}$. With clever bookkeeping, this can be done efficiently (c.f. Section B.2).

Once the local search terminates with a candidate $\hat{\mathcal{A}}(K)$, we seek to identify the best K from the grid. Lemma 3 shows that fixing K and $\hat{\mathcal{A}}(K)$, the optimization problem defining $U^{3DP}(\hat{\mathcal{A}}(K), K)$ (c.f. Problem (2.3)) is a convex stochastic optimization problem over $\mathbf{q}$, and hence, well-suited to a first-order optimization method. Thus, instead of comparing the objective values of the inner problem of Problem (5.6) to identify the best K , we run a first order method for each $(K, \hat{\mathcal{A}}(K))$ pair to evaluate the inner objective function of Problem (5.5) and compare these when choosing the best K . We summarize this heuristic in Algorithm 1. Experiments in Section 7 suggest Algorithm 1 already achieves near-optimal performance at a fraction of the MIO method's computational cost.

We next provide some details on how we implement projected stochastic gradient descent.

5.4. Evaluating $U^{3DP}(\mathcal{A}, K)$ via Projected SGD

For fixed $(\mathcal{A}, K)$, evaluating $U^{3DP}(\mathcal{A}, K)$ reduces to solving the convex stochastic program in Problem (4.5) over first-stage orders q . We use projected stochastic gradient descent (SGD), which is well suited to large-scale instances and settings where (D, s) are accessed through samples.

Each iteration requires (i) an unbiased subgradient of the objective with respect to q and (ii) projection onto the nonnegative orthant, which is available in closed form. The key computational step is evaluating the second-stage recourse function V^{3DP} and its subgradients, which can be computed efficiently in our setting due to the structural properties established below.

PROPOSITION 5 (Fractional-knapsack structure). *For any realization (D, s) , solving the recourse problem defining V^{3DP} is equivalent to solving a fractional knapsack problem that allocates limited 3DP capacity across components in $\mathcal{A}$ in decreasing order of per-unit printing margin.*

PROPOSITION 6 (Subgradients of V^{3DP}). *For any fixed $(\mathcal{A}, K)$ and realization (D, s) , the recourse problem defining V^{3DP} is convex. Moreover, given an optimal solution to the recourse problem, the subgradients of V^{3DP} admit explicit closed-form expressions.*

The explicit subgradient formulas and the corresponding fractional knapsack solution are stated and derived in the proofs in Section A.1. Together, Propositions 5 and 6 establish that V^{3DP} and its subgradients can be computed efficiently at each SGD iteration. This makes projected SGD a scalable method for repeatedly evaluating $U^{3DP}(\mathcal{A}, K)$ within the heuristic framework of Section 5.3.

6. Piloting 3DP as a Backup Strategy

In this section, we study a firm that has not yet adopted 3DP and is evaluating it as a backup strategy. In light of Proposition 1, we analyze adoption in two steps. First, we identify conditions under which backing up $\mathcal{A}_0$ alone, i.e., currently unprotected products, is cost-effective. Second, we examine when and why components should switch from dedicated to 3DP backup.

6.1. Backing Up Unprotected Products $\mathcal{A}_0$

Because firms are typically reluctant to modify existing dedicated backup contracts, we first evaluate the benefits of using 3DP solely to protect the previously unprotected components $\mathcal{A}_0$. In particular, we ask when there exists a 3DP capacity $K > 0$ that yields positive cost savings when just protecting $\mathcal{A}_0$, i.e., a K such that

$$U^{\text{tot-saved}}(\mathcal{A}_0, K) \equiv \sum_{j \in \mathcal{A}_0} U_j^0 - U^{3DP}(\mathcal{A}_0, K) - C^{3DP}(K) > 0. \quad (6.1)$$

Here, $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ denotes the total cost savings from backing up $\mathcal{A}_0$ with 3DP, and U_j^0 is the operational cost of component j when left unprotected. Since $C^{3\text{DP}}(K)$ is increasing and convex, and Lemma 3 shows that $K \mapsto U^{3\text{DP}}(\mathcal{A}_0, K)$ is convex, $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ is concave in K .

Moreover, because $C^{3\text{DP}}(0) = 0$, we have $U^{\text{tot-saved}}(\mathcal{A}_0, 0) = 0$: investing zero 3DP capacity is equivalent to leaving all components in $\mathcal{A}_0$ unprotected. Consequently, as K increases, $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ either decreases monotonically from zero or initially increases before eventually declining. In the latter case, any sufficiently small positive capacity yields cost savings. These two regimes can be distinguished by examining the derivative of $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ at $K = 0$.

THEOREM 6 (Necessary and Sufficient Conditions for 3DP Cost Savings under $\mathcal{A}_0$). *Let $C^{3\text{DP}}$ be increasing, convex, differentiable at 0 and $C^{3\text{DP}}(0) = 0$, and let $\hat{q}^0$ be the optimal first-stage order under no backup. Then there exists $K > 0$ such that $U^{\text{tot-saved}}(\mathcal{A}_0, K) > 0$ if and only if*

$$0 < \mathbb{E}_{D,s} \left[\max_{j \in \mathcal{A}_0} (v_j - c_j^{3\text{DP}}) \mathbb{I} \left\{ \left[D_j - \sum_{i \in \mathcal{S}_j} \hat{q}_{ij}^0 s_{ij} \right]^+ > 0 \right\} \right] - (C^{3\text{DP}})'(0). \quad (6.2)$$

If such a K exists, then $U^{\text{tot-saved}}(\mathcal{A}_0, K') > 0$ for all $K' \in (0, K]$.

Theorem 6 has an important managerial implication: When Eq. (6.2) holds, *any* sufficiently small investment in 3DP capacity delivers positive cost savings. This makes piloting 3DP a low-risk entry point: firms can start with minimal capacity, realize immediate benefits, and scale up as they accumulate operational experience.

6.2. Guidelines for Switching Components from Dedicated to 3DP Backup

We next examine when a firm should switch a component ℓ from dedicated backup to an existing 3DP portfolio $\mathcal{A}$ when operating with capacity K (e.g., the optimal capacity $K^*(\mathcal{A})$). The exact switching criterion compares the total cost of maintaining a dedicated backup against the incremental operational cost imposed on the 3DP system:

$$U_\ell^{\text{DB}} + C_\ell^{\text{DB}} > U^{3\text{DP}}(\mathcal{A} \cup \{\ell\}, K) - U^{3\text{DP}}(\mathcal{A}, K). \quad (6.3)$$

While (6.3) can be evaluated numerically, it offers limited intuition. To obtain interpretable decision rules, we approximate $U^{3\text{DP}}$ using the supermodular lower bound $L^{3\text{DP}}$ from Section 5.2, which yields the following sufficient approximation for switching:

$$U_\ell^0 - U_\ell^{\text{DB}} - C_\ell^{\text{DB}} < K \mathbb{E} \left[\left(r_\ell - \max_{j \in \mathcal{A}} r_j \mathbb{I} \{ \bar{D}_j^0 > 0 \} \right)^+ \mathbb{I} \{ \bar{D}_\ell^0 > 0 \} \right]. \quad (6.4)$$

Here, $\bar{D}_j^0 \equiv D_j - \sum_{i \in \mathcal{S}_j} \hat{q}_{ij}^0 s_{ij}$ denotes the shortfall of component j in the absence of backup, and $r_j \equiv v_j - c_j^{\text{3DP}}$ denotes the per-unit savings from using 3DP. The left-hand side captures the net benefit of using dedicated backup for ℓ rather than leaving it unprotected, while the right-hand side approximates the marginal value of adding ℓ to the existing 3DP portfolio, accounting for competition for limited capacity with components already in $\mathcal{A}$.

To obtain a simple necessary screening rule, we bound both sides of Eq. (6.4). For the right-hand side, the *most* that 3DP can save by adding ℓ is Kr_ℓ , corresponding to the best-case scenario in which all 3DP capacity is allocated to producing component ℓ . For the left-hand side, the benefit of covering ℓ with dedicated backup is at least the per-unit cost saved on each unit of shortfall, $v_\ell - c_\ell^{\text{DB}}$, times the expected shortfall, net of the fixed cost C_ℓ^{DB} . Although the expected shortfall under dedicated backup is endogenous, it is bounded below by $\mathbb{E}[\bar{D}_\ell^0]$, the expected shortfall when no backup is available. Combining these bounds yields the necessary condition

$$(v_\ell - c_\ell^{\text{DB}})\mathbb{E}[\bar{D}_\ell^0] - C_\ell^{\text{DB}} < Kr_\ell \iff \frac{v_\ell - c_\ell^{\text{DB}} - \kappa_\ell^{\text{DB}}}{r_\ell} < \frac{K}{\mathbb{E}[\bar{D}_\ell^0]}, \quad (6.5)$$

where $\kappa_\ell^{\text{DB}} \equiv C_\ell^{\text{DB}}/\mathbb{E}[\bar{D}_\ell^0]$ is the amortized fixed cost of dedicated backup.

To obtain a sufficient condition, we reverse the bounding logic. For the left-hand side, the operational benefit of dedicated backup is *at most* the per-unit savings $v_\ell - c_\ell^{\text{DB}}$ times the expected shortfall it covers under the dedicated-backup ordering policy. Let $\bar{D}_\ell^{\text{DB}} \equiv D_\ell - \sum_{i \in \mathcal{S}_\ell} \bar{q}_{i\ell}^{\text{DB}} s_{i\ell}$ denote this shortfall, where $\bar{q}^{\text{DB}}$ is the optimal first-stage order under dedicated backup. For the right-hand side, we construct a conservative lower bound on the marginal value of adding ℓ to 3DP by pessimistically increasing the per-unit savings of every component in $\mathcal{A}$ to $\max_{j \in \mathcal{A}} r_j$. This maximizes competition for capacity and reduces ℓ 's marginal advantage whenever any component in $\mathcal{A}$ is short. Combining these bounds yields the sufficient condition

$$(v_\ell - c_\ell^{\text{DB}})\mathbb{E}[\bar{D}_\ell^{\text{DB}}] - C_\ell^{\text{DB}} < K \left[\left(r_\ell - \max_{j \in \mathcal{A}} r_j \right)^+ \mathbb{P}(\exists j \in \mathcal{A} : \bar{D}_j^0 > 0 \mid \bar{D}_\ell^0 > 0) \right] \mathbb{P}(\bar{D}_\ell^0 > 0). \quad (6.6)$$

Taken together, (6.5)–(6.6) suggest five factors that govern whether a component is a strong candidate for switching to 3DP:

- **Cost advantage:** A large ratio $r_\ell/(v_\ell - c_\ell^{\text{DB}} - \kappa_\ell^{\text{DB}})$.
- **Shortfall likelihood:** A high probability $\mathbb{P}(\bar{D}_\ell^0 > 0)$.
- **Capacity intensity:** A small expected shortfall $\mathbb{E}[\bar{D}_\ell^0]$ relative to capacity K .
- **Portfolio dominance:** A large gap $r_\ell - \max_{j \in \mathcal{A}} r_j$.
- **Dependence with $\mathcal{A}$:** Low conditional probability $\mathbb{P}(\exists j \in \mathcal{A} : \bar{D}_j^0 > 0 \mid \bar{D}_\ell^0 > 0)$.

The first two factors depend only on marginal properties of component ℓ and correspond to the type of standalone screening rules emphasized in prior work on backup strategies (e.g., Saghafian and Van Oyen (2012)). Our analysis shows that such marginal criteria are incomplete in the presence of pooled, flexible backup capacity. The remaining factors are fundamentally portfolio-level: they measure whether ℓ diversifies the existing 3DP portfolio or instead congests it by forcing components to compete for scarce capacity in the same disruption scenarios. In the numerical studies that follow, we demonstrate that accounting for these portfolio effects yields meaningful improvements over marginal screening rules.

7. Empirical Case Study

We conduct a case study inspired by Mattel, a major toy company. Because Mattel's products are primarily plastic, we calibrate 3DP and sourcing costs using product weight. Accordingly, we replace the 3DP capacity constraint in Problem (2.4) with the weighted form

$$\sum_{j \in \mathcal{A}} w_j q_j^{3DP} \leq K,$$

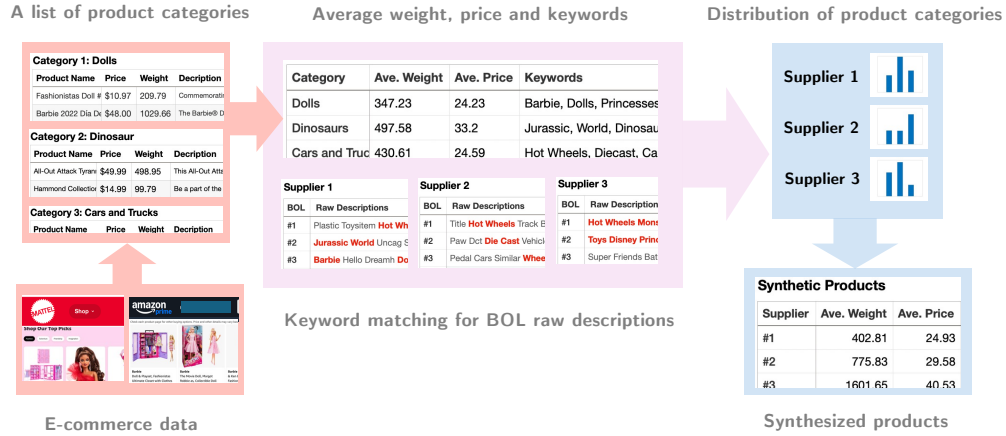
where w_j is the material weight (grams) required to print one unit of product j . We assume baseline primary-sourcing cost is proportional to weight at \$0.006 per gram (raw plastic plus a 20% markup). Because we do not observe proprietary supplier–component or component–product relationships, we assume a one-to-one correspondence among suppliers, components, and products in our experiments. As is standard in the literature, we interpret the objective in Problem (2.1) as regret relative to an oracle benchmark that meets all realized demand at baseline sourcing costs.

7.1. Calibration to Real-World Data

Because detailed operational data from Mattel are unavailable, we calibrate our model using public sources: (i) technical specifications of industrial 3DPs, (ii) product listings from Mattel.com and the Mattel Store on Amazon.com, and (iii) bill-of-lading (BOL) data on maritime imports. Figure 3 summarizes the calibration pipeline; additional details are provided in Section C.

7.1.1. 3DP Costs We model unit 3DP production cost as a multiple of the primary sourcing cost, setting $\kappa_j^{3DP} = \beta^{3DP} \kappa_j$ with a baseline $\beta^{3DP} = 2$, and report sensitivity for $\beta^{3DP} \in [1, 4]$. This range reflects the per-gram cost of non-metal printing materials (e.g., resin and PLA) together with a 50% infill assumption.

We assume a linear investment cost for 3DP capacity, $C^{3DP}(K) = c^{cap} K$, where $c^{cap} = Q^{3DP}/M^{3DP}$ is the unit capacity cost in dollars per gram. Here Q^{3DP} denotes the monthly depreciation cost per printer and M^{3DP} its monthly material output. We take a baseline printer cost of

Figure 3 Overview of the calibration procedure.

Note. We use e-commerce data from Mattel.com and the Mattel Store on Amazon.com to identify product categories, weights, prices, and descriptions. Category keywords are learned using TF-IDF scoring and applied to bill-of-lading (BOL) shipment descriptions to infer category mixes by supplier, yielding one synthetic product per supplier. Public printer specifications (Sintratec 2024) used to calibrate printing and capacity costs.

\$5,000 with a 10-year lifespan, yielding $Q^{3DP} = \$41.67$ per month (All3DP 2024). The output M^{3DP} is estimated from published printer specifications (nozzle width, layer thickness, movement speed, infill density, and operating hours); details are provided in Section C. This calibration yields $c^{cap} = \$0.0023$ per gram, and we report sensitivity to this value below.

7.1.2. Learning Product Categories We scrape Mattel.com and the Mattel Store on Amazon.com to identify seven product categories (e.g., dolls, vehicles, dinosaurs, etc.), along with each product's weight, sales price, and textual description. For each category ℓ , we compute the average unit weight w_ℓ^{ave} and average price κ_ℓ^{ave} , and extract a category-specific keyword list from product descriptions using TF-IDF scoring. These category averages and keywords are used in the next step to construct synthetic products from bill-of-lading data.

Bill of Lading	Carrier SCAC / Vessel Code (Name)	Voyage / Container Size / Type
XXXX333333333333	Eglv, Eagle Van Lines Inc / 3333333 (Ever Envoy)	3333x / 3000*333*333 / 33X3
Destination Port Name / Code	Company Name / Address	Arrival Date
Los Angeles / 2704	Mattel Import Services Corp / 333 Continental Blvd, El Segundo, CA, 90245, USA	XX/XX/20XX
Departure Port Name / Code	Supplier Name / Address	
Yantian / 57078	Shenzhen Hutchison Inland / Container Depots Co Ltd Xintian Hutchison Warehouse Distripark Huanguan Rd Sth G1 Town Baoan Disct Sz O/B	
HS Code / Full Description	Product Description	Quantity (Unit) / TEU / Weight
950611 / Toys, games and sports requisites; parts and accessories ...	American Girl Dolls Plastic Toys Dinotrux Diecast Hot Wheels Heavy Etc Market Mdm	333 (CTN) / 33 / 3333 kilogram

Table 1 Sample bill of lading. One such report is submitted to the Federal Customs and Border Protection Agency for every import by a US Firm. As highlighted, these forms provide partial information about Tier 1 suppliers, types of goods imported, and, demand for that supplier.

7.1.3. Identifying Suppliers and Constructing Synthetic Products from Bill-of-Lading Data

A bill of lading (BOL) is a shipping record filed with U.S. Customs and Border Protection that documents the shipper, consignee, goods description, and shipment weight for each maritime import. We obtain BOL data from ImportYeti.com, which aggregates filings across firms and years, and focus on 55 suppliers with more than 100 shipments to Mattel over the past decade. These suppliers account for the majority of import volume and form the basis of our experiments.

In principle, BOL data could be used to directly infer product-level demand, but two features limit their granularity (see Table 1). First, a supplier typically ships multiple products, yet BOL descriptions are often generic (e.g., “plastic toys”), without product-level identifiers or quantities. Second, shipments reliably report total weight but rarely report unit counts. As a result, we cannot disaggregate shipments into individual products.

To address these limitations, we construct a single *synthetic product* for each supplier. This construction implicitly assumes that disruptions at a supplier proportionately affect all products sourced from that supplier, and that products across suppliers are not substitutable.

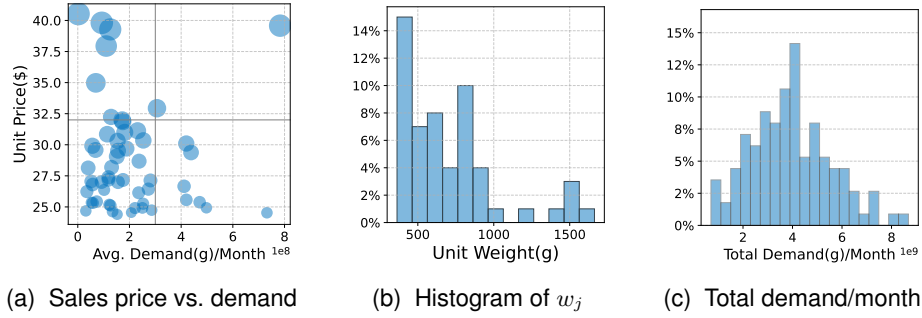
We leverage the product categories learned in the previous subsection. For each supplier j and category ℓ , let b_j^ℓ denote the fraction of that supplier’s BOLs whose descriptions contain at least one keyword from category ℓ . These proportions approximate the supplier’s category mix. We then define the synthetic product’s unit weight and sales price as

$$w_j = \sum_{\ell} b_j^\ell w_{\ell}^{\text{ave}}, \quad \kappa_j^{\text{sell}} = \sum_{\ell} b_j^\ell \kappa_{\ell}^{\text{ave}}.$$

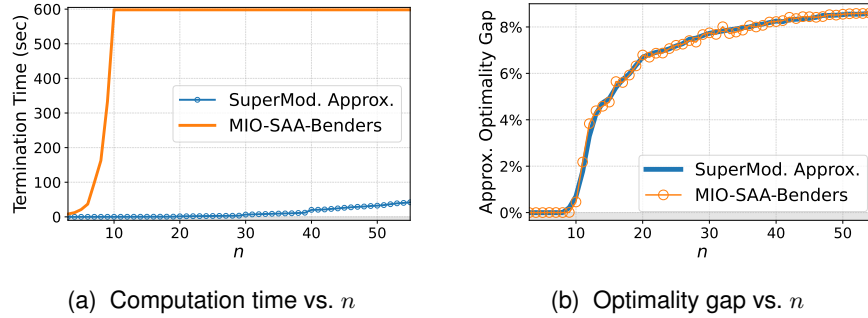
Monthly demand is inferred by aggregating shipment weights and converting them to unit counts using w_j . To simplify computation, each empirical demand distribution is discretized via K -means clustering into a three-point distribution. Demand is assumed independent across suppliers; with 55 suppliers, this already yields more than 10^{26} joint demand realizations.

Although somewhat involved, we do believe this procedure yields a reasonable approximation to the supply chain for a company like Mattel. Figure 4 gives an overview of the demand and price of these synthetic products.

To complete our calibration, we require yield data for each of these 55 suppliers. Yield data are rarely shared externally due to their proprietary nature. Thus, we model random yield explicitly as $s_j \in \{\alpha_j, 1\}$, where $1 - \alpha_j \in (0, 1)$ and $p_j = \mathbb{P}(s_j = 1 - \alpha_j)$ are assumed known. By default, $\alpha_j = 0.05$, $p_j = 0.05$ and s_j ’s are independent, unless specified otherwise.

Figure 4 Synthetic Products Calibrated to BOL Data.

Note. Our calibration yields 55 synthetic products, one per supplier. Panel (a) shows a positive correlation between price and demand (marker size proportional to per-unit weight), with many low-volume, low-cost products. Panel (b) highlights substantial heterogeneity in weight (and thus cost). Panel (c) reports aggregate firm demand, which exhibits significant variability.

Figure 5 Comparison of supermodular approximation and MIO for optimizing $\mathcal{A}$ 

7.2. How Accurate is the SuperMod Approx Heuristic?

We evaluate the performance of the SuperMod Approx heuristic by comparing it to the mixed-integer formulation in Theorem 4 on instances of increasing size. Because the number of scenarios grows exponentially in n , direct solution quickly becomes impractical. For benchmarking, we therefore compare SuperMod Approx to a mixed-integer sample-average approximation (MIO-SAA), where expectations are approximated using 5% of all possible scenarios and the solver runtime is capped at 600 seconds.

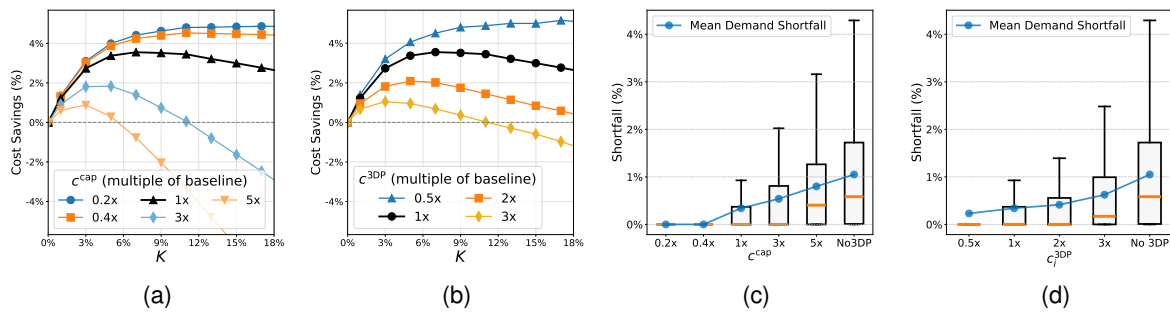
Figure 5 summarizes the results. As shown in Fig. 5a, MIO-SAA becomes computationally intractable once $n \geq 10$, while SuperMod Approx scales gradually up to the full instance with $n = 55$ suppliers. To assess solution quality at larger scales, Fig. 5b reports optimality gaps relative to the best Benders lower bound available at termination. Across all n , SuperMod Approx achieves gaps comparable to or smaller than those of MIO-SAA at a fraction of the computational cost. From a managerial perspective, the proposed heuristic delivers near-optimal 3DP portfolio decisions at scales suitable for practical resilience planning in large, disaggregated supply chains.

7.3. How much value does 3DP offer?

Of course, a key question is if 3DP provides any cost savings over traditional back up strategies. In Panels (a) and (b) of Fig. 6 we plot the cost savings relative to a system without 3DP as a function of the amount of capacity K purchased. As our theory suggests, the optimal K is a relatively small fraction of the maximum demand, but the savings are modest. For our baseline parameters, at the optimal K , we see a savings of just under 4%, but, at larger costs of capacity or costs of printing, this benefit quickly disappears.

Perhaps more striking are panels (c) and (d) of Fig. 6. Here we plot the distribution of the demand shortfall, i.e., the total unmet demand after utilizing the received primary orders and backup productions. Note, this quantity is random depending on the random realizations of D_j and s_j for all j . We compute these boxplots using 10^5 random samples of the vector $((D_j, s_j) : j = 1, \dots, 55)$. Here we see that although cost improvements are modest, 3DP effectively reduces the amount of unmet demand, particularly in tail scenarios. This benefit persists even for large costs of capacity or printing. In many ways we see this as the primary argument for 3DP as a resilience strategy; although average benefits are small, the additional flexibility helps guard against large vales of unmet demand.

Figure 6 Cost savings and demand shortfalls: varying c^{cap} and c^{3DP}



Note. Panels (a) and (b): Cost savings (%) relative to a system with dedicated backup but no 3DP. Savings are modest, and the optimal K is a relatively small fraction of demand. Panels (c) and (d) display box plots of demand shortfalls. Here we see a more striking reduction in the tails of the distribution. The x-axes indicate multiples of the baseline c^{cap} and c_j^{3DP} . Box plots labeled “No 3DP” correspond to the system with dedicated backups only and no 3DP.

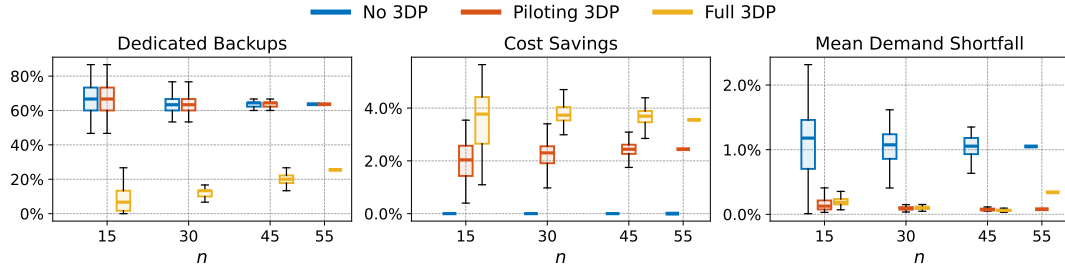
Figure 7 decomposes the benefits of introducing 3DP by comparing three strategies: a system with dedicated backups only, a “piloting” strategy that backs up previously unprotected products using 3DP, and the optimal 3DP policy that may also switch some dedicated backups. We evaluate these strategies on 100 random subsets with $n \in \{15, 30, 45\}$ suppliers and on the full system ($n = 55$).

The left panel shows that, under the optimal policy, a nontrivial fraction of products initially backed up by dedicated resources are switched to 3DP. The middle panel shows that the piloting

strategy yields more than half of the total cost savings, while the right panel shows also achieves most of the reduction in average unmet demand.

These results suggest a phased adoption of 3DP may be sufficient: deploying 3DP first to cover previously unprotected products yields many of the benefits without necessitating restructuring of existing resilience strategies.

Figure 7 Backup switch after introducing 3DP.



Note. From left to right: percentage of n assigned to dedicated backups, cost savings (relative to the cost without 3DP) and mean demand shortfall (relative to max demand) for $n \in \{15, 30, 45, 55\}$. We compare the dedicated backup-only strategy (blue), the piloting 3DP strategy (red, backing up only unprotected products), and the full 3DP strategy (yellow, potentially switching backups).

7.4. Impact of disruption correlations

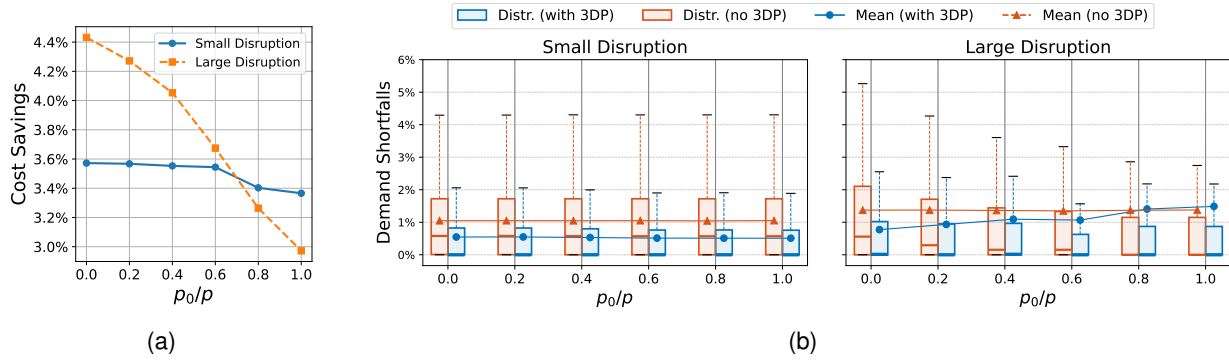
Theorem 3 shows that 3DP performs worst relative to dedicated backups when demand shortfalls are comonotonic. We empirically explore this effect by studying how the value of 3DP changes as disruptions become more correlated.

We introduce a latent global disruption factor $X_0 \sim \text{Bernoulli}(p_0)$ with $p_0 \leq 0.05$, together with idiosyncratic factors $X_j \sim \text{Bernoulli}\left(\frac{p-p_0}{1-p_0}\right)$ that are independent across suppliers. Supplier yield is given by $s_j = 1 - \alpha \max(X_j, X_0)$, so that each supplier fails with marginal probability $p = 0.05$. By varying the ratio p_0/p from 0 to 1, we interpolate smoothly from independent to fully comonotonic disruptions. We consider both small ($\alpha = 0.05$) and large ($\alpha = 1$) yield-loss regimes.

Figure 8 reports cost savings and demand shortfalls as correlation increases. Under small yield losses, cost savings decline only modestly as disruptions become more correlated. Under large yield losses, the decline is more pronounced, though 3DP continues to offer some cost savings even in the comonotonic limit. Demand shortfalls are consistently reduced by 3DP across correlation levels, but under large yield losses this benefit diminishes as correlation increases, since simultaneous failures can overwhelm limited 3DP capacity.

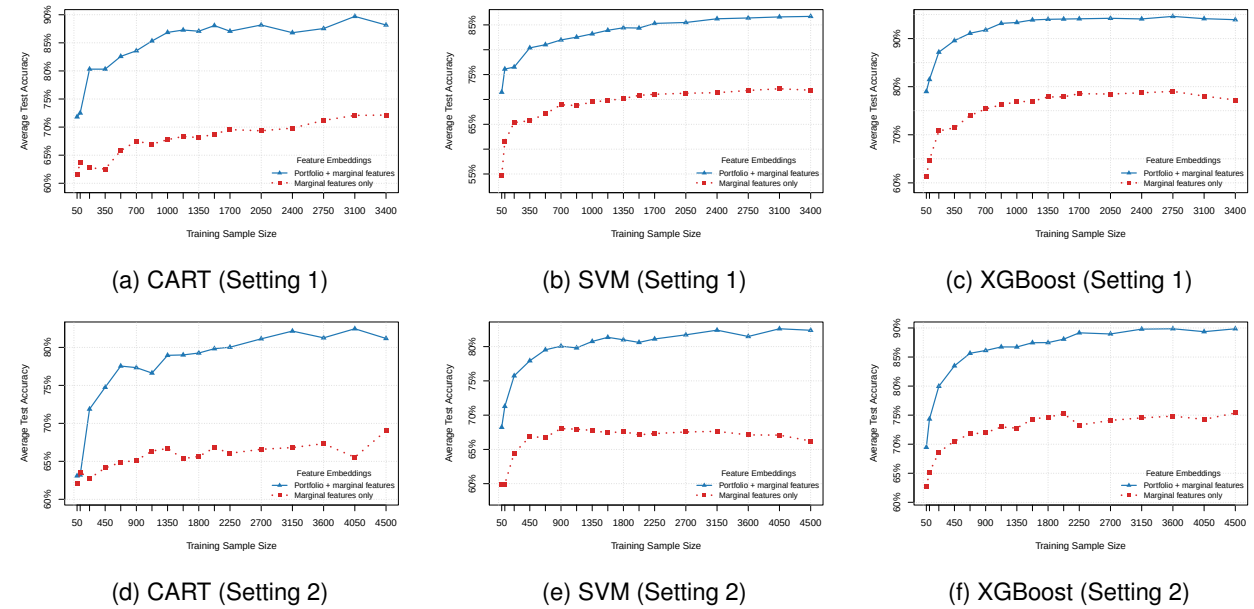
Overall, stronger disruption correlations weaken the effectiveness of 3DP as a resilience strategy, particularly when yield losses are large. However, even under highly correlated disruptions, 3DP can still provide meaningful reductions in demand shortfalls and modest cost benefits.

Figure 8 Cost savings and demand shortfalls: varying correlations among disruptions



Note. Panel (a): Cost savings (relative to no 3DP) as p_0/p increases from 0 (independent s) to 1 (comonotonic s). Panel (b): Demand shortfall (relative to total demand) vs. p_0/p . Both panels compare 5% (“Small”) and 100% (“Large”) yield loss disruption regimes.

Figure 9 Out-of-sample accuracy for predicting backup switching (from dedicated to 3DP).



Note. Features: marginal-only (red curves) vs. portfolio-augmented (blue curves).

7.5. What kinds of components are the best candidates for 3DP backup?

As shown in Section 6.2, switching a product from dedicated backup to 3DP depends not only on marginal characteristics of that product, but also on how it interacts with the existing 3DP portfolio. To quantify the importance of these portfolio effects, we use a machine learning approach that predicts backup-switching decisions from engineered features and compares predictive performance with and without portfolio features.

Feature sets. We compare two feature embeddings. The *marginal-only* embedding includes the marginal drivers from Section 6.2 (e.g., cost advantage and shortfall likelihood), together with basic marginal attributes such as service levels. The *portfolio-augmented* embedding adds the portfolio

features from Section 6.2—capacity intensity, portfolio dominance, and dependence with $\mathcal{A}$ —which capture interactions between a candidate product and the current 3DP set.

Data-generating mechanism. We generate labeled data by repeatedly sampling and solving synthetic instances with $n = 10$ components. For each instance, we draw disruption parameters $p \sim \text{Uniform}(0, 1)$ and $p_0 \sim \text{Uniform}(0, p)$ (as in the latent-factor model in the previous subsection), solve for the optimal 3DP set $\mathcal{A}^*$ and the initial unprotected set $\mathcal{A}_0$, and then construct a *reference* portfolio $\mathcal{A} = \mathcal{A}_0 \cup \hat{\mathcal{A}}^*$, where $\hat{\mathcal{A}}^*$ is a random 50% subset of $\mathcal{A}^* \setminus \mathcal{A}_0$. For each $\ell \notin \mathcal{A}$, we create a labeled data point indicating whether ℓ would be switched to 3DP when evaluated relative to $\mathcal{A}$ at capacity $K^*(\mathcal{A})$, together with both feature embeddings computed relative to $\mathcal{A}$.

We consider two settings. *Setting 1* repeats this procedure on 800 independently sampled instances. *Setting 2* isolates the role of dependence by sampling 300 base instances and generating three variants. The three variants differ only in correlation strength – drawing p_0 uniformly from $[0, \frac{1}{3}p]$, $[\frac{1}{3}p, \frac{2}{3}p]$, and $[\frac{2}{3}p, p]$, respectively – yielding 900 total instances. In this controlled setting, marginal features are identical across variants, but portfolio features and switching labels differ.

Learning results. For each setting and feature embedding, we train classifiers on increasing training sizes and evaluate out-of-sample accuracy; implementation details and model specifications are reported in the appendix. As shown in Fig. 9, incorporating portfolio features consistently improves predictive performance relative to marginal-only features. In particular, under the controlled-correlation setting, test accuracy increases from approximately 75% to 90% when portfolio features are included.

Taken together, these results provide empirical evidence that portfolio features play a central role in backup-switching decisions. Beyond marginal product characteristics, interactions with the existing backup portfolio should therefore be considered when qualitatively assessing whether a product is a strong candidate for 3DP backup. This provides a data-driven screening tool for identifying promising 3DP candidates beyond simple marginal heuristics.

8. Conclusion

In this paper we explore the possibility of using 3DP a flexible backup resource as part of a larger portfolio of supply chain resilience strategies. Doing so requires solving a host of operational problems around which products to backup with 3DP, how much 3DP capacity to acquire, and how to structure both primary orders from suppliers and recourse printing actions. To that end, we formulate a mixed-binary stochastic optimization problem describing the setting, and derive several properties of its optimal solution that help assess the potential value and costs of adopting a 3DP strategy.

We also provide a scalable heuristic for solving this problem based on constructing a supermodular approximation. Through an empirical case study inspired by Mattel, we establish that with current technology, 3DP offers only a modest savings over dedicated backup (3-4%), but that it offers a much more significant reduction in the amount of unmet demand in the system. For risk-sensitive firms or settings where qualitative branding risks are serious, such benefits might justify the investment.

There are a number of interesting direction for extension. We have considered a single firm and its Tier 1 suppliers. One could instead consider a supply-chain of multiple firms, each one transforming input components into output components passed to another layer, and ask which firms in this network would benefit most from 3DP backups, and for which components. Moreover, we have taken a deliberately conservative viewpoint, neglecting potential benefits of 3DP from co-location with demand. Modeling these features appropriately might reveal additional benefits.

Overall, 3DP and additive manufacturing have long been considered inviable for large-scale manufacturing and supply chain, and relegated to low-volume applications like prototyping. While there are very real practical challenges to widespread 3DP adoption including quality assurance, upskilling a workforce, and navigating regulatory requirements, there are also very real advantages to incorporating flexible backup strategies in a firm's supply-chain resilience plan. We hope our work inspires academics to think more broadly about the potential of this technology.

9. Data Availability and Reproducibility

All code for reproducing our experiments can be found at https://anonymous.4open.science/r/3DP_resilience_experiments-5FE1/. In the spirit of reproducibility of research, we also provide the complete dataset of 55 suppliers with details on their synthetic products within this repository. Data obtained from [ImportYeti.com](https://importyeti.com) is proprietary and available for purchase. We cannot share this data directly. Interested researchers should contact this firm.

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Appendix A: Omitted Proofs

In this section, we present the proofs of the key results discussed in this paper.

A.1. Proofs of “Workhorse” Results: Propositions 5 and 6 and Lemma 6

This section provides the proofs of three key results that will be used repeatedly in the subsequent analysis: (i) the fractional knapsack equivalence of V^{3DP} (Proposition 5), which yields an efficient evaluation subroutine; (ii) the closed-form subgradient expression (Proposition 6); and (iii) a characterization of minimal-magnitude solutions for univariate convex problems (Lemma 6 stated below), which facilitates the analysis of the optimal 3DP capacity and first-stage order. Given their central role in the upcoming proofs, we present their formal derivations here for ease of reference.

Proof of Proposition 5 We begin with a reformulation of $V^{3DP}(\mathbf{y}, K)$:

$$V^{3DP}(\mathbf{y}, K) = \sum_{j \in \mathcal{A}} v_j y_j^+ + h_j y_j^- + f^{3DP}(\mathbf{y}^+, K), \quad (\text{A.1})$$

where $\mathbf{y}^+$ is the vector consists of demand shortfall y_j^+ 's, and function f^{3DP} is defined by

$$\begin{aligned} f^{3DP}(\mathbf{z}, K) &\equiv \min_{\mathbf{q}^{3DP}} \sum_{j \in \mathcal{A}} -(v_j - c_j^{3DP}) q_j^{3DP} \\ \text{s.t. } &0 \leq q_j^{3DP} \leq z_j, \forall j \in \mathcal{A} \quad \text{and} \quad \sum_{j \in \mathcal{A}} q_j^{3DP} \leq K. \end{aligned} \quad (\text{A.2})$$

This reformulation results from the following manipulations:

- For all $j \in \mathcal{A}$ such that $y_j < 0$, there is no demand shortfall to handle, hence the corresponding $q_j^{3DP} = 0$ in Problem (2.4). Thus, $[y_j - q_j^{3DP}]^-$ equals y_j^- at optimality.
- For all $j \in \mathcal{A}$ such that $y_j \geq 0$, it is never optimal to use 3DP to back up more than the demand shortfall y_j . Therefore, $0 \leq q_j^{3DP} \leq y_j^+$, and $[y_j - q_j^{3DP}]^+$ equals $y_j^+ - q_j^{3DP}$.

The first two terms of the summand in Eq. (A.1) do not depend on $\mathbf{q}^{3DP}$. This decision variable only occurs in Problem (A.2), which is a fractional knapsack problem. $\square$

Following Eq. (A.1), the optimal $\mathbf{q}^{3DP}$ for Problem (2.4) can be computed via the standard fractional knapsack solution procedure (Algorithm 2). Hereafter, we assume indices in $\mathcal{A}$ are pre-sorted in descending order of $v_j - c_j^{3DP}$.

Algorithm 2 Optimal $\mathbf{q}^{3DP}$ for V^{3DP}

Require: Demand gap y_j for all $j \in \mathcal{A}$ and capacity K . The remaining capacity K' is initially set to K .

- 1: **for** $j = 1 \dots |\mathcal{A}|$ **do**
 - 2: Set $\bar{q}_j^{3DP} = \min\{K', y_j^+\}$, and update K' as $K' - \bar{q}_j^{3DP}$.
 - 3: **return** $\bar{\mathbf{q}}^{3DP}$ as the optimal solution for Problem (2.4) which defines $V^{3DP}(\mathbf{y}, K)$
-

In other words, the optimal fulfillment $\bar{q}_j^{3DP}$ of demand shortfalls y_j^+ is obtained by iteratively filling y_j^+ in descending order of the per unit fulfillment reward $v_j - c_j^{3DP}$, until all shortfalls are addressed or the capacity K is exhausted.

Proof of Proposition 6 As indicated in Proposition 6, the subgradients of V^{3DP} admit closed-form expressions. We derive them here by leveraging the structure of the optimal solution $\bar{\mathbf{q}}^{3DP}$ from Algorithm 2. Since these expressions play a key role in subsequent proofs, we encapsulate their precise statement in Lemma 5 below for ease of reference.

LEMMA 5 (Convexity and Subgradients of V^{3DP}). *The following properties hold for V^{3DP} :*

(i) $V^{3DP}(\mathbf{y}, K)$ is jointly convex in $\mathbf{y}$ and K .

(ii) Let $\bar{\mathbf{q}}^{3DP}$ be the output of Algorithm 2 for given $\mathbf{y}$ and K , and define $j^* \equiv \min\{j \in \mathcal{A} : y_j > 0, \bar{q}_j^{3DP} < y_j\}$ as the first positive shortfall that is not fully filled. Then, a subgradient λ of $V^{3DP}(\mathbf{y}, K)$ with respect to K is given by

$$\lambda = \begin{cases} -(v_{j^*} - c_{j^*}^{3DP}) & \text{if } \sum_{j \in \mathcal{A}} y_j^+ > K \\ 0 & \text{otherwise.} \end{cases} \quad (\text{A.3})$$

Moreover, a subgradient μ_j of $V^{3DP}(\mathbf{y}, K)$ with respect to y_j is given by

$$\mu_j = \begin{cases} -h_j & \text{if } y_j \leq 0, \\ v_j & \text{if } y_j > 0 \text{ and } \bar{q}_j^{3DP} < y_j, \\ c_j^{3DP} - \lambda & \text{otherwise.} \end{cases} \quad (\text{A.4})$$

Proof of Lemma 5 This proof relies on an epigraphical reformulation of $V^{3DP}(\mathbf{y}, K)$:

$$\begin{aligned} \min_{\mathbf{q}^{3DP} \geq 0, \zeta^+, \zeta^-} \quad & \sum_{j \in \mathcal{A}} (c_j^{3DP} q_j^{3DP} + v_j \zeta_j^+ + h_j \zeta_j^-) \\ \text{s.t.} \quad & \sum_{j \in \mathcal{A}} q_j^{3DP} \leq K, \end{aligned} \quad (\text{A.5a})$$

$$y_j - q_j^{3DP} \leq \zeta_j^+, \quad 0 \leq \zeta_j^+, \quad \forall j \in \mathcal{A}, \quad (\text{A.5b})$$

$$-y_j + q_j^{3DP} \leq \zeta_j^-, \quad 0 \leq \zeta_j^-, \quad \forall j \in \mathcal{A}. \quad (\text{A.5c})$$

where we have replaced $[y_j - q_j^{3DP}]^+$ and $[y_j - q_j^{3DP}]^-$ in the objective of Problem (2.4) with nonnegative variables ζ_j^+ and ζ_j^- , subject to constraints $y_j - q_j^{3DP} \leq \zeta_j^+$ and $y_j - q_j^{3DP} \geq -\zeta_j^-$. The dual of this optimization problem is:

$$\begin{aligned} \max_{\lambda \leq 0, \mu^+, \mu^-} \quad & K\lambda + \mathbf{y}^\top (\boldsymbol{\mu}^- - \boldsymbol{\mu}^+) \\ \text{s.t.} \quad & \lambda - \mu_j^+ + \mu_j^- \leq c_j^{3DP}, \end{aligned} \quad \forall j \in \mathcal{A}, \quad (\text{A.6a})$$

$$-v_j \leq \mu_j^+ \leq 0, \quad \forall j \in \mathcal{A}, \quad (\text{A.6b})$$

$$-h_j \leq \mu_j^- \leq 0, \quad \forall j \in \mathcal{A}. \quad (\text{A.6c})$$

Thus, $V^{3DP}(\mathbf{y}, K)$ is the maximum of linear functions and jointly convex in $\mathbf{y}, K$, proving statement (i).

To prove statement (ii), we will apply Danskin's theorem (Proposition 4.5.1 in Bertsekas et al. (2003)), which provides the subgradients of V^{3DP} in terms of an optimal solution to Problem (A.6). Hence, we first construct a primal and dual pair of optimal solutions to Problems A.5 and A.6. Namely,

- Let $\mathbf{q}^{3DP} = \bar{\mathbf{q}}^{3DP}$ where $\bar{\mathbf{q}}^{3DP}$ is the output of Algorithm 2.
- Let $\zeta_j^+ = [y_j - \bar{q}_j^{3DP}]^+$ and $\zeta_j^- = [y_j - \bar{q}_j^{3DP}]^-$, for all $j \in \mathcal{A}$.
- Let λ be defined as in Eq. (A.3).
- For all $j \in \mathcal{A}$ such that $y_j \leq 0$, set $\mu_j^+ = 0$ and $\mu_j^- = -h_j$.
- For all $j \in \mathcal{A}$ such that $y_j > 0$ and $\bar{q}_j^{3DP} = y_j$, set $\mu_j^+ = \lambda - c_j^{3DP}$ and $\mu_j^- = 0$.
- For all $j \in \mathcal{A}$ such that $y_j > 0$ and $\bar{q}_j^{3DP} < y_j$, set $\mu_j^+ = -v_j$ and $\mu_j^- = 0$.

Since $\bar{\mathbf{q}}^{3DP}$ is optimal for Problem (2.4), it follows that $(\mathbf{q}^{3DP}, \boldsymbol{\zeta}^+, \boldsymbol{\zeta}^-)$ are optimal to Problem (A.5).

To check dual feasibility, we consider each $j \in \mathcal{A}$ following the cases outlined above:

- For all $j \in \mathcal{A}$ such that $y_j \leq 0$: μ_j^+ and μ_j^- satisfy Eqs. (A.6b) and (A.6c) by construction. They also satisfy Eq. (A.6a), because $\lambda \leq 0$, and hence, the left side of Eq. (A.6a) is non-positive while the right side is non-negative.

- For all $j \in \mathcal{A}$ such that $y_j > 0$ and $\bar{q}_j^{3DP} = y_j$: Eq. (A.6c) is satisfied by construction. Since $\bar{q}_j^{3DP} = y_j$, product j cannot have been added to the knapsack after j^* in Algorithm 2. Hence, $v_j - c_j^{3DP} \geq v_{j^*} - c_{j^*}^{3DP} \geq -\lambda$. Rearranging shows that $-v_j \leq \lambda - c_j^{3DP}$, which implies the left inequality in Eq. (A.6b). The right inequality follows immediately because $\lambda \leq 0$ and $c_j^{3DP} \geq 0$. Finally, by substitution, the Eq. (A.6a) holds with equality.
- For all $j \in \mathcal{A}$ such that $y_j > 0$ and $\bar{q}_j^{3DP} < y_j$: Eqs. (A.6b) and (A.6c) hold by construction. Moreover, because $\bar{q}_j^{3DP} < y_j$, we have j could not have been added to the knapsack before j^* in Algorithm 2, and so $v_j - c_j^{3DP} \leq v_{j^*} - c_{j^*}^{3DP}$, which proves the Eq. (A.6a) holds.

Thus, our constructed dual solution is dual feasible.

It remains to show complementary slackness:

$$\left(\sum_{j \in \mathcal{A}} q_j^{3DP} - K \right) \lambda = 0, \quad (-q_j^{3DP} - \zeta_j^+ + y_j) \mu_j^+ = 0, \quad (-q_j^{3DP} + \zeta_j^- + y_j) \mu_j^- = 0, \quad \forall j \in \mathcal{A}, \quad (\text{A.7})$$

$$(\lambda - \mu_j^+ + \mu_j^- - c_j^{3DP}) q_j^{3DP} = 0, \quad (-\mu_j^+ - v_j) \zeta_j^+ = 0, \quad (-\mu_j^- - h_j) \zeta_j^- = 0, \quad \forall j \in \mathcal{A}.$$

These constraints can again be checked by cases in exactly the same manner as above for dual feasibility, which in turn confirms that $\mu_j^+, \mu_j^-, \lambda$, constitute a dual optimal solution.

Hence, by Danskin's theorem, subgradients with respect to y_j and K are given by $\mu_j = \mu_j^- - \mu_j^+$ and λ , respectively, concluding the proof. $\square$

REMARK 1. We note that the subgradient of V^{3DP} is unique except when $(\mathbf{y}, K)$ falls into one of the following scenarios: (i) $\sum_{j \in \mathcal{A}} y_j^+ > K$ and $\bar{q}_{j^*}^{3DP} = y_{j^*}$; or (ii) $\sum_{j \in \mathcal{A}} y_j^+ = K$. In these cases, the optimal λ for the dual problem Problem (A.6) forms an interval, and we choose its upper bound as the λ presented in Lemma 5. Accordingly, in the third case of Eq. (A.4), this choice of λ makes $c_j^{3DP} - \lambda$ the smallest μ_j under this scenario.

This choice of λ and μ_j allows us to directly apply Eq. (A.4) and Eq. (A.3) to derive the right derivatives of $K \mapsto V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, K)$ and $q_j \mapsto V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, K)$, which we use in the proofs of Theorem 1 and Theorem 2 to characterize the optimal 3DP capacity and first-stage order. $\square$

Recall that in Theorem 1 and Theorem 2, ties among multiple optima are resolved by selecting the one with the smallest ℓ_1 -norm. When the decision variable is a positive scalar, the following lemma characterizes the least ℓ_1 -norm solution using only the right derivative of the objective function.

LEMMA 6 (Minimal Solution of Univariate Convex Function). *Let $f : \mathbb{R} \mapsto \mathbb{R}$ be convex and let $\bar{x} \equiv \min\{x \geq 0 : f'_+(x) \geq 0\}$, where $f'_+(x)$ denotes the right derivative of f . Then, $\bar{x}$ is the smallest optimal solution to $\min_{x \geq 0} f(x)$.*

Proof of Lemma 6 We first note that $\bar{x} = \min\{x \geq 0 : f'_+(x) \geq 0\}$ is well-defined because f'_+ is right-continuous (Bertsekas et al. 2003, Prop. 4.1.1(e)). Since any optimal solution x^* must satisfy $f'_+(x^*) \geq 0$ (first-order optimality condition), we immediately have $x^* \geq \bar{x}$ by the minimality of $\bar{x}$. Thus, it suffices to show that $\bar{x}$ itself is optimal. If $\bar{x} = 0$, optimality follows directly from the boundary condition $f'_+(0) \geq 0$. If $\bar{x} > 0$, then optimality requires $f'_-(\bar{x}) \leq 0 \leq f'_+(\bar{x})$, where f'_- denotes the left derivative. Suppose, for contradiction, that $f'_-(\bar{x}) > 0$. By the left-continuity of f'_- (Bertsekas et al. 2003, Prop. 4.1.1(e)), there exists $0 < x_0 < \bar{x}$ such that $f'_-(x_0) > 0$. Since $f'_+(x_0) \geq f'_-(x_0)$ (Bertsekas et al. 2003, Prop. 4.1.1(a)), it follows that $f'_+(x_0) > 0$. This contradicts the minimality of $\bar{x}$, completing the proof. $\square$

A.2. Proofs from Section 2

Proof of Lemma 2. First, consider the objective in Problem (2.2), which defines U_j^{DB} :

$$\mathbb{E} [c_{ij}q_{ij}s_{ij} + c_j^{\text{DB}}[D_j - q_{ij}s_{ij}]^+ + h_j[D_j - q_{ij}s_{ij}]^-]. \quad (\text{A.8})$$

This function is convex in q_{ij} since both $[D_j - q_{ij}s_{ij}]^+$ and $[D_j - q_{ij}s_{ij}]^-$ are convex and expectation preserves convexity. By Lemma 6, the smallest optimal solution $\bar{q}_{ij}^{\text{DB}}$ is the smallest q_{ij} for which the right derivative is nonnegative.

Differentiating under the expectation (justified by uniform Lipschitz continuity), the right derivative equals

$$\mathbb{E} [c_{ij}s_{ij} - c_j^{\text{DB}}s_{ij}\mathbb{I}\{D_j > q_{ij}s_{ij}\} + h_js_{ij}\mathbb{I}\{D_j \leq q_{ij}s_{ij}\}].$$

Rearranging shows that this derivative is nonnegative if and only if

$$\mathbb{E} [s_{ij}\mathbb{I}\{D_j \leq q_{ij}s_{ij}\}] \geq \left(\frac{c_j^{\text{DB}} - c_{ij}}{c_j^{\text{DB}} + h_j} \right) \mathbb{E} [s_{ij}].$$

Hence $\bar{q}_{ij}^{\text{DB}} = \bar{q}_{ij}(c_j^{\text{DB}})$ as defined in Eq. (2.5). $\square$

Proof of Lemma 3. Throughout this proof, let $\mathcal{A}$ be fixed. **Part (i).** Since $V^{\text{3DP}}(\mathbf{y}, K)$ is jointly convex in $(\mathbf{y}, K)$ (Lemma 5), its composition with the affine mapping $\mathbf{q} \mapsto \mathbf{D} - \mathbf{q} \circ \mathbf{s}$ ensures that $V^{\text{3DP}}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, K)$ remains jointly convex in $(\mathbf{q}, K)$. **Part (ii).** As the expectation operator preserves convexity (Shapiro et al. 2021, Thm. 7.46), the objective in Problem (2.3) is convex. Given the convexity of the feasible region, the result follows. **Part (iii).** By Part (i) and the linearity of expectation, the objective function is jointly convex in $(\mathbf{q}, K)$. Since partial minimization over a convex set preserves convexity (Bertsekas et al. 2003, Prop. 2.3.6), $U^{\text{AM}}(\mathcal{A}, K)$ is convex in K . $\square$

A.3. Proofs from Section 3

Proof of Proposition 1 To establish $\mathcal{A}_0 \subseteq \mathcal{A}^*$, it suffices to show that the optimal $\mathcal{A}$ for any fixed $K \geq 0$ includes $\mathcal{A}_0$. This holds because adding an unprotected product to $\mathcal{A}$ and choosing to never print it in the second stage is costless.

To show that the inclusion can be strict, it suffices to construct an example in which $\mathcal{A} = \mathcal{N}$ is strictly better than $\mathcal{A} = \mathcal{A}_0$. Consider a setting where $0 \leq D_j \leq \bar{D}_j$ almost surely for all $j \in \mathcal{N}$, and $C^{\text{3DP}}(K) = 0$. Then it is optimal to set $K = \sum_{j \in \mathcal{A}} \bar{D}_j$. Let U_j^0 and U_j^∞ denote the optimal operational costs of component j under zero and infinite 3DP capacity, respectively. Under $\mathcal{A} = \mathcal{N}$, the total cost equals $\sum_{j \in \mathcal{N}} U_j^\infty$, whereas under $\mathcal{A} = \mathcal{A}_0$ the total cost equals

$$\sum_{j \in \mathcal{A}_0} U_j^0 + \sum_{j \in \mathcal{A}_0^c} (U_j^{\text{DB}} + C_j^{\text{DB}}).$$

Hence $\mathcal{A} = \mathcal{N}$ is strictly better than $\mathcal{A} = \mathcal{A}_0$ if

$$\sum_{j \in \mathcal{A}_0^c} U_j^\infty < \sum_{j \in \mathcal{A}_0^c} (U_j^{\text{DB}} + C_j^{\text{DB}}),$$

which can be ensured by choosing C_j^{DB} sufficiently large. Finally, to guarantee $\mathcal{A}_0 \neq \mathcal{N}$, it suffices that for some j ,

$$U_j^{\text{DB}} + C_j^{\text{DB}} < U_j^0.$$

Since neither U_j^{DB} nor U_j^∞ depends on the stockout cost v_j under uncapacitated backup, increasing v_j raises U_j^0 without affecting the other terms. Thus we can ensure $\mathcal{A}_0 \neq \mathcal{N}$. This constructs an example in which $\mathcal{A} = \mathcal{A}_0$ is not optimal, so the optimal $\mathcal{A}$ can strictly include $\mathcal{A}_0$. $\square$

Proof of Theorem 1 Let $j \in \mathcal{A}$ and $i \in \mathcal{S}_j$ be fixed. Recall that $q^*(\mathcal{A}, K)$ is an optimal solution to Problem (2.3) (which defines $U^{3DP}(\mathcal{A}, K)$) with the minimal ℓ_1 -norm. We first note that $q_{ij}^*(\mathcal{A}, K)$ is the smallest optimal solution to Problem (2.3) when minimizing only over $q_{ij} \geq 0$, while keeping $q_{\ell j} = q_{\ell j}^*(\mathcal{A}, K)$ fixed for all $\ell \in \mathcal{S}_j \setminus \{i\}$ and $q_{\ell k} = q_{\ell k}^*(\mathcal{A}, K)$ for all $\ell \in \mathcal{S}_k, k \in \mathcal{A} \setminus \{j\}$, i.e., $q_{ij}^*(\mathcal{A}, K)$ solves the following univariate optimization problem:

$$\begin{aligned} \min_{q \geq 0} \quad & c_{ij}q\mathbb{E}[s_{ij}] + \sum_{k \in \mathcal{A} \setminus \{j\}} \sum_{\ell \in \mathcal{S}_k} c_{\ell k}q_{\ell k}^*(\mathcal{A}, K)\mathbb{E}[s_{\ell k}] + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} c_{\ell j}q_{\ell j}^*(\mathcal{A}, K)\mathbb{E}[s_{\ell k}] \\ & + \mathbb{E} \left[c_{ij}qs_{ij} + V^{3DP} \left(\left[D_k - \sum_{\ell \in \mathcal{S}_k} \hat{q}_{\ell k}s_{\ell k} \right]_{k \in \mathcal{A}}, \mathcal{A}, K \right) \right], \end{aligned}$$

which is equivalent to

$$\min_{q \geq 0} \quad \mathbb{E} \left[c_{ij}qs_{ij} + V^{3DP} \left(\left[D_k - \sum_{\ell \in \mathcal{S}_k} \hat{q}_{\ell k}s_{\ell k} \right]_{k \in \mathcal{A}}, \mathcal{A}, K \right) \right]. \quad (\text{A.9})$$

In both problems, we let

$$\hat{q}_{\ell k} \equiv \begin{cases} q_{\ell k}^*(\mathcal{A}, K) & \text{if } k \neq j \text{ or } k = j \text{ but } \ell \in \mathcal{S} \setminus \{i\} \\ q & \text{otherwise.} \end{cases} \quad (\text{A.10})$$

For ease of notation, we also define the following demand shortfall vector:

$$D - \hat{q} \circ s \equiv \left[D_k - \sum_{\ell \in \mathcal{S}_k} \hat{q}_{\ell k}s_{\ell k} \right]_{k \in \mathcal{A}}.$$

Note that we must have $q_{ij}^*(\mathcal{A}, K)$ being the minimal solution to the univariate Problem (A.9). Indeed, if this were not the case, we could use it to construct an optimal solution to Problem (2.3) with a smaller ℓ_1 -norm than $q^*(\mathcal{A}, K)$, contradicting its definition.

Now, since Problem (A.9) is a univariate convex optimization problem, by Lemma 6, $q_{ij}^*(\mathcal{A}, K)$ is the smallest $q \geq 0$ for which the right derivative of its objective function is nonnegative. This right derivative is given by:

$$\begin{aligned} & \frac{d}{dq^+} \mathbb{E} [c_{ij}qs_{ij} + V^{3DP}(D - \hat{q} \circ s, \mathcal{A}, K)] \\ &= \mathbb{E} \left[c_{ij}s_{ij} + \frac{d}{dq^+} V^{3DP}(D - \hat{q} \circ s, \mathcal{A}, K) \right] \end{aligned} \quad (\text{A.11a})$$

$$= \mathbb{E} [c_{ij}qs_{ij} - s_{ij}\mu_j(D, s, q)] \quad (\text{A.11b})$$

$$= \mathbb{E} \left[c_{ij}s_{ij} + h_j s_{ij} \mathbb{I} \left\{ D_j \leq qs_{ij} + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \hat{q}_{\ell j}s_{\ell j} \right\} - s_{ij}\mu_j(D, s, q) \mathbb{I} \left\{ D_j > qs_{ij} + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \hat{q}_{\ell j}s_{\ell j} \right\} \right], \quad (\text{A.11c})$$

where $\mu_j(D, s, q)$ denotes the μ_j from Eq. (A.4) after setting $y = D - \hat{q} \circ s$ in Lemma 5. Specifically, Eq. (A.11a) swaps differentiation and integration since the integrand is uniformly Lipschitz in q with parameter at most $c_{ij} + v_j + h_j$. Eq. (A.11b) applies Proposition 4.2.2 of Bertsekas et al. (2003), which states that the right derivative of $q \mapsto V^{3DP}(D - \hat{q} \circ s, \mathcal{A}, K)$ is its largest subgradient. By the chain rule, this subgradient is given by $-s_{ij}\mu_j(D, s, q)$, since $\mu_j(D, s, q)$ is constructed as the smallest possible subgradient of $y_j \mapsto V^{3DP}(y, \mathcal{A}, K)$ at $y = D - \hat{q} \circ s$ (see Remark 1). Finally, Eq. (A.11c) follows from the specific construction of $\mu_j(D, s, q)$ as given in Eq. (A.4).

In summary, we now have the following characterization of $q_{ij}^*(\mathcal{A}, K)$

$$q_{ij}^*(\mathcal{A}, K) = \min \{q \geq 0 : H(q) \geq 0\}, \quad (\text{A.12})$$

where

$$H(q) \equiv \mathbb{E} \left[c_{ij} s_{ij} + h_j s_{ij} \mathbb{I} \left\{ D_j \leq q s_{ij} + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \hat{q}_{\ell j} s_{\ell j} \right\} - s_{ij} \mu_j(\mathbf{D}, \mathbf{s}, q) \mathbb{I} \left\{ D_j > q s_{ij} + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \hat{q}_{\ell j} s_{\ell j} \right\} \right]. \quad (\text{A.13})$$

Recall that $\bar{q}_{ij}^0 = \bar{q}_{ij}(v_j, D_j)$ and $\bar{q}_{ij}^\infty = \bar{q}_{ij}(c_j^{3\text{DP}}, D_j - \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \bar{q}_{\ell j}^0 s_{\ell j})$. By the definition of $\bar{q}_{ij}(\cdot, \cdot)$ in Eq. (2.5), $\bar{q}_{ij}^0$ and $\bar{q}_{ij}^\infty$ satisfy the following conditions after algebraic rearrangement:

$$\bar{q}_{ij}^0 = \min\{q \geq 0 : H^0(q) \geq 0\} \quad \text{and} \quad \bar{q}_{ij}^\infty = \min\{q \geq 0 : H^\infty(q) \geq 0\}, \quad (\text{A.14})$$

where

$$H^0(q) \equiv \mathbb{E} [c_{ij} s_{ij} + h_j s_{ij} \mathbb{I} \{D_j \leq q s_{ij}\} - s_{ij} v_j \mathbb{I} \{D_j > q s_{ij}\}]. \quad (\text{A.15})$$

$$H^\infty(q) \equiv \mathbb{E} \left[c_{ij} s_{ij} + h_j s_{ij} \mathbb{I} \left\{ D_j \leq q s_{ij} + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \bar{q}_{\ell j}^0 s_{\ell j} \right\} - s_{ij} c_j^{3\text{DP}} \mathbb{I} \left\{ D_j > q s_{ij} + \sum_{\ell \in \mathcal{S}_j \setminus \{i\}} \bar{q}_{\ell j}^0 s_{\ell j} \right\} \right] \quad (\text{A.16})$$

Now, to establish $\bar{q}_{ij}^\infty \leq q_{ij}^{3\text{DP}}(\mathcal{A}, K) \leq \bar{q}_{ij}^0$, it suffices to show that for all $\mathbf{D}, \mathbf{s}$ and $q \geq 0$ such that $D_j > q s_{ij} + \sum_{\ell \in \mathcal{S}_j} \hat{q}_{\ell j} s_{\ell j}$, the following inequalities hold:

$$c_j^{3\text{DP}} \leq \mu_j(\mathbf{D}, \mathbf{s}, q) \leq v_j. \quad (\text{A.17})$$

To see why Eq. (A.17) suffices, note that if it holds, then for all $q \geq 0$ we first have $H^0(q) \leq H(q)$ holds since the first-stage orders and yields are all non-negative valued. Consequently, we obtain $0 \leq H^0(\bar{q}_{ij}^0) \leq H(\bar{q}_{ij}^0)$, where the first inequality follows from Eq. (A.12). In other words, $q = \bar{q}_{ij}^0$ satisfies the inequality $H(q) \geq 0$. Since $q_{ij}^*(\mathcal{A}, K)$ is the smallest such q (cf. Eq. (A.12)), it follows that $q_{ij}^*(\mathcal{A}, K) \leq \bar{q}_{ij}^0$. Combining this result with the bound $c_j^{3\text{DP}} \leq \mu_j(\mathbf{D}, \mathbf{s}, q)$ from Eq. (A.17), we establish that $H(q) \leq H^\infty(q)$. By an analogous argument, it follows that $q_{ij}^*(\mathcal{A}, K) \geq \bar{q}_{ij}^\infty$.

Thus, the proof is complete once we establish Eq. (A.17) for all $\mathbf{D}, \mathbf{s}$ and $q \geq 0$ with $D_j > q s_{ij} + \sum_{\ell \in \mathcal{S}_j} \hat{q}_{\ell j} s_{\ell j}$. We verify this claim under the following cases, where we set $\bar{q}^{3\text{DP}} \mathbf{y} = \mathbf{D} - \hat{\mathbf{q}} \circ \mathbf{s}$ for simplicity, and j^* are defined as in Lemma 5 under this $\mathbf{y}$:

- i) If $\bar{q}_j^{3\text{DP}} < y_j$, then from Eq. (A.4), $\mu_j(\mathbf{D}, \mathbf{s}, q) = v_j$, satisfying Eq. (A.17).
- ii) If $\bar{q}_j^{3\text{DP}} = y_j$ and $\sum_{j \in \mathcal{A}} y_j^+ \leq K$, then from Eq. (A.4), $\mu_j(\mathbf{D}, \mathbf{s}, q) = c_j^{3\text{DP}}$, satisfying Eq. (A.17).
- iii) Finally, when $\bar{q}_j^{3\text{DP}} = y_j$ and $\sum_{j \in \mathcal{A}} y_j^+ > K$, Eq. (A.4) gives $\mu_j(\mathbf{D}, \mathbf{s}, q) = c_j^{3\text{DP}} + (v_{j^*} - c_{j^*}^{3\text{DP}})$, hence we have $\mu_j(\mathbf{D}, \mathbf{s}, q) \geq c_j^{3\text{DP}}$ since $v_{j^*} - c_{j^*}^{3\text{DP}} \geq 0$. Moreover, in this scenario, shortfall j is fully filled, so by construction, $j^* > j$. Since $v_j - c_j^{3\text{DP}} \leq v_{j^*} - c_{j^*}^{3\text{DP}}$, it follows that $\mu_j(\mathbf{D}, \mathbf{s}, q) \leq v_j$ after rearranging terms.

In summary, we have shown that Eq. (A.17) holds in all cases, therefore proving $\bar{q}_{ij}^\infty \leq q_{ij}^*(\mathcal{A}, K) \leq \bar{q}_{ij}^0$. $\square$

Proof of Theorem 2 For ease of reference, let $r_j \equiv v_j - c_j^{3\text{DP}}$ and denote the objective function of Problem (2.3) (which defines $U^{3\text{DP}}$) as $F^{3\text{DP}}(K)$. We begin by characterizing $K^*(\mathcal{A}, q)$, which is key to proving statements (i) and (ii).

Recall that $K^*(\mathcal{A}, q)$ is the smallest optimal capacity for a given $\mathcal{A}$ and a fixed first-stage order $q \geq 0$, making it the smallest minimizer of $C^{3\text{DP}}(K) + F^{3\text{DP}}(K)$ for $K \geq 0$, where $C^{3\text{DP}}(K) = c^{\text{cap}} K$. Since this is a univariate convex problem, it follows from Lemma 6 that

$$K^*(\mathcal{A}, q) = \min\{K \geq 0 : H^*(K, q) \geq 0\}, \quad (\text{A.18})$$

where $H^*(K, \mathbf{q})$ is the right derivative of $c^{\text{cap}}K + F^{\text{3DP}}(K)$. We derive $H^*(K, \mathbf{q})$ in closed form as follows:

$$H^*(K, \mathbf{q}) = \frac{d}{dK^+} (c^{\text{cap}}K + F^{\text{3DP}}(K)) = c^{\text{cap}} + \frac{d}{dK^+} \mathbb{E} [V^{\text{3DP}}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, K)] \quad (\text{A.19a})$$

$$= c^{\text{cap}} + \mathbb{E} \left[\frac{d}{dK^+} V^{\text{3DP}}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, K) \right] \quad (\text{A.19b})$$

$$= c^{\text{cap}} + \mathbb{E} [\lambda(\mathbf{D}, \mathbf{s}, K)] \quad (\text{A.19c})$$

$$= c^{\text{cap}} - \mathbb{E} \left[r_{j^*} \mathbb{I} \left\{ \sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^+ > K \right\} \right]. \quad (\text{A.19d})$$

In (A.19b), $\lambda(\mathbf{D}, \mathbf{s}, K)$ is the λ from Eq. (A.3), defined under $\mathbf{y} = \mathbf{D} - \mathbf{q} \circ \mathbf{s}$; the index j^* in (A.19d) is defined in Lemma 5 under the same setting. Equation (A.19a) holds because V^{3DP} is the only term in F^{3DP} that depends on K . In Eq. (A.19b), differentiation and integration are interchanged, which is valid since the integrand is uniformly Lipschitz in q with a parameter at most $\max_{j \in \mathcal{N}} (v_j - c_j^{\text{3DP}})$. Next, Eq. (A.19b) applies Proposition 4.2.2 of Bertsekas et al. (2003), which states that the right derivative of $K \mapsto V^{\text{3DP}}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, K)$ is its largest subgradient. By construction (see Remark 1), this subgradient is precisely $\lambda(\mathbf{D}, \mathbf{s}, K)$, as given by λ in Eq. (A.4), whose explicit form yields Eq. (A.19d).

To prove part (i) of the theorem, we substitute $v_j - c_j = r$ for all $j \in \mathcal{A}$ into Eq. (A.19), which simplifies Eq. (A.18) to

$$K^*(\mathcal{A}, \mathbf{q}) = \inf \left\{ K \geq 0 : c^{\text{cap}} - r \mathbb{P} \left(\sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^+ > K \right) \geq 0 \right\} \quad (\text{A.20})$$

$$= \inf \left\{ K \geq 0 : \mathbb{P} \left(\sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^+ > K \right) \leq \frac{c^{\text{cap}}}{r} \right\}. \quad (\text{A.21})$$

By the definition, $K^*(\mathcal{A}, \mathbf{q})$ is thus the $(1 - \frac{c^{\text{cap}}}{r})$ -quantile of $\sum_{j \in \mathcal{A}} [D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$, thereby proving part (i).

We now prove part (ii) of the theorem, where r_j values are heterogeneous. Recall that $K^*(\mathcal{A})$ is the smallest optimal capacity under $\mathcal{A}$, so we have $K^*(\mathcal{A}) = K^*(\mathcal{A}, \mathbf{q}^*(\mathcal{A}))$, where $\mathbf{q}^*(\mathcal{A})$ is the optimal first-stage order with the minimal ℓ_1 -norm. For convenience, we define the following functions

$$H^{\min}(K, \mathbf{q}) \equiv c^{\text{cap}} - r^{\min} \mathbb{P} \left(\sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^+ > K \right) \quad \text{and} \quad H^{\max}(K, \mathbf{q}) \equiv c^{\text{cap}} - r^{\max} \mathbb{P} \left(\sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij} \right]^+ > K \right).$$

Since $r^{\min} \leq r_{j^*} \leq r^{\max}$ almost surely, it follows that

$$H^{\min}(K, \mathbf{q}) \leq H^*(K, \mathbf{q}) \leq H^{\max}(K, \mathbf{q}). \quad (\text{A.22})$$

The proof of part (ii) is completed as follows:

$$\left(1 - \frac{c^{\text{cap}}}{r^{\max}}\right)\text{-quantile of } \sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^{\infty} s_{ij} \right]^+ \leq \left(1 - \frac{c^{\text{cap}}}{r^{\max}}\right)\text{-quantile of } \sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij}^*(\mathcal{A}) s_{ij} \right]^+ \quad (\text{A.23a})$$

$$= \inf \left\{ K \geq 0 : \mathbb{P} \left(\sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij}^*(\mathcal{A}) s_{ij} \right]^+ > K \right) \leq \frac{c^{\text{cap}}}{r^{\max}} \right\} \quad (\text{A.23b})$$

$$= \inf \{ K \geq 0 : H^{\max}(K, \mathbf{q}^*(\mathcal{A})) \geq 0 \} \quad (\text{A.23c})$$

$$\leq \inf \{ K \geq 0 : H^*(K, \mathbf{q}^*(\mathcal{A})) \geq 0 \} \quad (\text{A.23d})$$

$$\leq \inf \{K \geq 0 : H^{\min}(K, \mathbf{q}^*(\mathcal{A})) \geq 0\} \quad (\text{A.23e})$$

$$= \inf \left\{ K \geq 0 : \mathbb{P} \left(\sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij}^*(\mathcal{A}) s_{ij} \right]^+ > K \right) \leq \frac{c^{\text{cap}}}{r^{\min}} \right\} \quad (\text{A.23f})$$

$$= \left(1 - \frac{c^{\text{cap}}}{r^{\min}} \right) \text{-quantile of } \sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} q_{ij}^*(\mathcal{A}) s_{ij} \right]^+ \quad (\text{A.23g})$$

$$\leq \left(1 - \frac{c^{\text{cap}}}{r^{\min}} \right) \text{-quantile of } \sum_{j \in \mathcal{A}} \left[D_j - \sum_{i \in \mathcal{S}_j} \bar{q}_{ij}^0 s_{ij} \right]^+. \quad (\text{A.23h})$$

The inequalities in (A.23a) and (A.23h) follow from the monotonicity of the quantiles of $\sum_{j \in \mathcal{A}} [D_j - \sum_{i \in \mathcal{S}_j} q_{ij} s_{ij}]^+$ in q_{ij} and the bounds $\bar{q}_{ij}^\infty \leq q_{ij}^*(\mathcal{A}) \leq \bar{q}_{ij}^0$ in Theorem 1. The identities in (A.23b) and (A.23g) are standard quantile representations, and (A.23c) and (A.23f) follow by rewriting these representations in terms of $H^{\max}$ and $H^{\min}$.

For (A.23d) and (A.23e), note that each side is an infimum over K with the same objective. By (A.22), we have $H^*(K, \mathbf{q}^*(\mathcal{A})) \leq H^{\max}(K, \mathbf{q}^*(\mathcal{A}))$ and $H^{\min}(K, \mathbf{q}^*(\mathcal{A})) \leq H^*(K, \mathbf{q}^*(\mathcal{A}))$ for all K , so the feasible set enlarges when replacing $H^{\max}$ by H^* and enlarges again when replacing H^* by $H^{\min}$, yielding the stated inequalities. Finally, the right-hand side of (A.23d) equals $K^*(\mathcal{A}, \mathbf{q}^*(\mathcal{A}))$, and since $K^*(\mathcal{A}) = K^*(\mathcal{A}, \mathbf{q}^*(\mathcal{A}))$, statement (ii) is thus proven. $\square$

Proof of Theorem 3 In this proof, we fix the marginal distributions of $(\mathbf{D}, \mathbf{s})$ as well as $\mathcal{A}, K$, and $\mathbf{q}$, omitting them from the notation. Without loss of generality, we assume the indices in $\mathcal{A}$ are ordered in descending $r_j \equiv v_j - c_j^{\text{3DP}}$.

The key idea of the proof is to show that $f^{\text{3DP}}(\mathbf{z})$ in Problem (A.2) is supermodular on $\mathbb{R}_+^m$, i.e., for all $\mathbf{z}, \bar{\mathbf{z}} \geq 0$,

$$f^{\text{3DP}}(\mathbf{z}) + f^{\text{3DP}}(\bar{\mathbf{z}}) \leq f^{\text{3DP}}(\min\{\mathbf{z}, \bar{\mathbf{z}}\}) + f^{\text{3DP}}(\max\{\mathbf{z}, \bar{\mathbf{z}}\}),$$

where $\min\{\mathbf{z}, \bar{\mathbf{z}}\}$ and $\max\{\mathbf{z}, \bar{\mathbf{z}}\}$ denote the elementwise minimum and maximum, respectively.

To see why supermodularity of f^{3DP} suffices, note that by Goovaerts and Dhaene (1999, Theorem 6), among random vectors $\mathbf{y}^+ \equiv [\mathbf{D} - \mathbf{q} \circ \mathbf{s}]^+$ with fixed marginal distributions, $\mathbb{E}[f^{\text{3DP}}(\mathbf{y}^+)]$ reaches its maximum when $\mathbf{y}^+$ is comonotonic. Since $\mathbb{E}[f^{\text{3DP}}(\mathbf{y}^+)]$ is the only term in the objective of Problem (2.1) that depends on the correlation of $\mathbf{y}^+$, the overall optimal value is maximized under comonotonicity. It therefore remains to establish that f^{3DP} is supermodular.

By Corollary 2.6.1 in Topkis (1998), it suffices to show that $f^{\text{3DP}}(\mathbf{z})$ has *increasing differences*. Specifically, for any $i, j \in \mathcal{A}$, fix z_ℓ for all $\ell \in \mathcal{A} \setminus \{i, j\}$ and express f^{3DP} solely as a function of z_i and z_j , i.e., $f^{\text{3DP}}(z_i, z_j)$, then

$$f^{\text{3DP}}(\bar{z}_i, z'_j + \Delta) - f^{\text{3DP}}(\bar{z}_i, z'_j) \geq f^{\text{3DP}}(z'_i, z'_j + \Delta) - f^{\text{3DP}}(z'_i, z'_j), \quad (\text{A.24})$$

for all nonnegative scalars $z'_i, \bar{z}_i, z'_j$ and Δ with $z'_i \leq \bar{z}_i$.

We first prove Eq. (A.24) for sufficiently small $\Delta > 0$ such that $\lambda(z'_i, z'_j) = \lambda(z'_i, z'_j + \Delta)$, where $\lambda(z_i, z_j)$ is the optimal λ in the following dual reformulation of $f^{\text{3DP}}(z_i, z_j)$, with z_ℓ fixed for all $\ell \notin \{i, j\}$:

$$\begin{aligned} f^{\text{3DP}}(z_i, z_j) &= \max_{\lambda \leq 0, \mu} K\lambda + \sum_{\ell \in \mathcal{A}} \mu_\ell z_\ell &= \max_{\lambda \leq 0} K\lambda - \sum_{\ell \in \mathcal{A}} [r_\ell + \lambda]^+ z_\ell. \\ \text{s.t. } &\mu_\ell \leq -\lambda - r_\ell, \mu_\ell \leq 0, \forall \ell \in \mathcal{A} \end{aligned} \quad (\text{A.25})$$

The second equality in Problem (A.25) follows from the dual constraint $\mu_\ell \leq \min\{0, -\lambda - r_\ell\} = -[r_\ell + \lambda]^+$, hence maximizing the dual objective yields $\mu_\ell = -[r_\ell + \lambda]^+$ at optimality.

By the same argument in Lemma 5, after replacing y_ℓ^+ with z_ℓ , the λ defined in Eq. (A.3) remains optimal for Problem (A.25). Thus, applying Danskin's theorem, the subgradient of $z_j \mapsto f^{3DP}(z_i, z_j)$ is given by $-[r_j + \lambda(z_i, z_j)]^+$, and:

$$f^{3DP}(z'_i, z'_j + \Delta) - f^{3DP}(z'_i, z'_j) \leq -\Delta[r_j + \lambda(z'_i, z'_j + \Delta)]^+ = -\Delta[r_j + \lambda(z'_i, z'_j)]^+, \quad (\text{A.26})$$

$$f^{3DP}(\bar{z}_i, z'_j + \Delta) - f^{3DP}(\bar{z}_i, z'_j) \geq -\Delta[r_j + \lambda(\bar{z}_i, z'_j)]^+. \quad (\text{A.27})$$

The inequalities in Eq. (A.26) and Eq. (A.27) follow from subgradient inequality of convex function f^{3DP} , while the equality in Eq. (A.26) holds due to the $\lambda(z'_i, z'_j) = \lambda(z'_i, z'_j + \Delta)$ assumption. Since $t \mapsto [t]^+$ is non-decreasing, for Eq. (A.24) to hold, it suffices to show that $\lambda(\bar{z}_i, z'_j) \leq \lambda(z'_i, z'_j)$.

Recall that the optimal λ for Problem (A.25) (as given in Eq. (A.3)) is either $-r_{j^*}$, where j^* is the first positive shortfall that is not fully filled, or 0 when all shortfalls are fully filled. To unify these two scenarios, we introduce an artificial shortfall $z_{m+1} = K$ with reward $r_{m+1} = 0$, so that we can always write the optimal λ as $-r_{j^*}$. Under this framework, the index j^* is smaller in the $z_i = \bar{z}_i$ case since it requires filling more shortfalls than $z_i = z'_i$. Consequently, $-r_{j^*}$ is smaller for $z_i = \bar{z}_i$, leading to $\lambda(\bar{z}_i, z'_j) \leq \lambda(z'_i, z'_j)$. Therefore, Eq. (A.24) holds under $\lambda(z'_i, z'_j) = \lambda(z'_i, z'_j + \Delta)$.

To extend this result to general $\Delta > 0$, note that $z_j \mapsto f^{3DP}(z)$ is convex piecewise affine from Problem (A.25). Thus, the interval $[z'_j, z'_j + \Delta]$ can be partitioned into subintervals $[z'_j + \Delta_{\ell-1}, z'_j + \Delta_\ell]$ for $\ell = 1, \dots, L$, where $\Delta_0 = 0$ and $\Delta_L = \Delta$, such that within the interior of each subinterval, $\lambda(z'_i, z_j)$ remains constant and takes a unique value λ_ℓ^* .

Although $\lambda(z'_i, z'_j + \Delta_{\ell-1})$ and $\lambda(z'_i, z'_j + \Delta_\ell)$ are not unique, Bertsekas et al. (2003)[Proposition 4.2.3 (b)] ensures that both can take the value λ_ℓ^* . Thus, when restricted to each subinterval $[z'_j + \Delta_{\ell-1}, z'_j + \Delta_\ell]$, we may assume without loss of generality that $\lambda(z'_i, z'_j + \Delta_{\ell-1}) = \lambda(z'_i, z'_j + \Delta_\ell)$ and apply Eq. (A.24) under this condition to conclude that

$$f^{3DP}(z'_i, z'_j + \Delta_\ell) - f^{3DP}(z'_i, z'_j + \Delta_{\ell-1}) \leq f^{3DP}(\bar{z}_i, z'_j + \Delta_\ell) - f^{3DP}(\bar{z}_i, z'_j + \Delta_{\ell-1}). \quad (\text{A.28})$$

Finally, summing Eq. (A.28) over all $\ell = 1, \dots, L$ establishes Eq. (A.24) for any $\Delta > 0$, thus completing the proof. $\square$

A.4. Proofs from Section 4

Proof of Lemma 4 Recall that $y = D - q \circ s$ for fixed (D, s) and $q \geq 0$. Without loss of generality, we set $q_{ij} = 0$ whenever $x_j = 0$, since all terms indexed by such j vanish in Problems 4.1 and 4.2. Moreover, for $x_j = 1$, we restrict $\bar{q}_{ij}^\infty \leq q_{ij} \leq \bar{q}_{ij}^0$, as optimal solutions lie in this interval by Theorem 1.

Let $\hat{V}^{3DP}$ denote the right-hand side of (4.3). To prove $V^{3DP} = \hat{V}^{3DP}$, it suffices to show that, with M_j^1, M_j^2, M_j^3 defined in Eq. (4.4), the optimal solution $\bar{q}^{3DP}$ of Problem (4.2) is feasible for $\hat{V}^{3DP}$ and that the objectives coincide.

When $x_j = 1$, constraints (4.3b), (4.3c), and (4.3d) reduce to

$$0 \leq q_j^{3DP} \leq M_j^1, \quad z_j^{3DP} = q_j^{3DP}, \quad |z_j^{3DP} - y_j| \leq M_j^3. \quad (\text{A.29})$$

Since $\bar{q}_j^{3DP} \leq [D_j - \sum_{i \in S_j} q_{ij} s_{ij}]^+$ and D_j, s_{ij} are bounded, the definitions of M_j^1 and M_j^3 ensure these inequalities are satisfied by $\bar{q}_j^{3DP}$.

When $x_j = 0$, the same constraints reduce to

$$q_j^{3DP} = 0, \quad z_j^{3DP} = y_j, \quad |z_j^{3DP}| \leq M_j^2. \quad (\text{A.30})$$

Note that when $x_j = 0$, the optimal $\bar{q}_j^{3DP}$ for V^{3DP} is arbitrary and can be set to 0, trivially satisfying (A.30). It remains to verify that $|z_j^{3DP}| \leq M_j^2$. Since $z_j^{3DP} = y_j = D_j - \sum_{i \in S_j} q_{ij} s_{ij}$, choosing $M_j^2 = M_j^3$ is sufficient for $|z_j^{3DP}| \leq M_j^2$ to hold.

Combining (A.29) and (A.30), constraint (4.3e) in $\widehat{V}^{3DP}$ reduces to $\sum_{j:x_j=1} q_j^{3DP} \leq K$, which is equivalent to $\sum_{j \in \mathcal{N}} q_j^{3DP} x_j \leq K$ in V^{3DP} . Hence any optimal $\bar{q}^{3DP}$ for V^{3DP} is feasible for $\widehat{V}^{3DP}$. Substituting the same simplifications into the objective of Problem (4.3) yields exactly the objective of Problem (4.2), and therefore $\widehat{V}^{3DP} = V^{3DP}$.

Convexity of $\widehat{V}^{3DP}(\mathbf{y}, \mathbf{x})$ follows by penalizing the constraints in (4.3) with an indicator function and applying Proposition 2.3.6 of Bertsekas et al. (2003). □

Proof of Theorem 4 By Lemma 4 and Theorem 1, it holds that (here $V^{3DP}(\mathbf{y}, \mathbf{x}, K)$ is given by Problem (4.3)):

$$\begin{aligned} U^{3DP}(\mathbf{x}, K) = \min_{\mathbf{q}} \quad & \sum_{j=1}^n c_{ij} q_{ij} \mathbb{E}[s_{ij}] + \mathbb{E}_{D,s} [V^{3DP}(\mathbf{D} - \mathbf{q} \circ \mathbf{s}, \mathbf{x}, K)] \\ \text{s.t.} \quad & \bar{q}_{ij}^\infty x_j \leq q_{ij} \leq \bar{q}_{ij}^0 x_j, \quad \forall j = 1 \dots n. \end{aligned} \quad (\text{A.31})$$

This establishes the equivalence between Problem (4.5) and Problem (2.1). Moreover, the objective function in Problem (4.5) is convex, as V^{3DP} is convex (Lemma 4) and the expectation operator preserves this property. □

A.5. Proofs from Section 5

Proof of Proposition 2. First, note that $\sum_{j \in \mathcal{A}} \sum_{i \in \mathcal{S}_j} c_{ij} q_{ij} \mathbb{E}[s_{ij}]$ is modular in $\mathcal{A}$, and since expectation preserves supermodularity, it suffices to show that V^{3DP} is supermodular in $\mathcal{A}$. Using the reformulation in Eq. (A.1), this reduces to proving that $f^{3DP}(\mathbf{z}, \mathcal{A}, K)$ is supermodular in $\mathcal{A}$ for any $\mathbf{z} \in \mathbb{R}_+^n$ and $K \geq 0$, where f^{3DP} is given in Problem (A.2) and explicitly written here as a function of $\mathcal{A}$. Specifically, for all $\mathcal{S}, \mathcal{T} \subseteq \mathcal{N}$ with $\mathcal{S} \subset \mathcal{T}$ and $i \notin \mathcal{T}$, we aim to show that:

$$f^{3DP}(\mathbf{z}, \mathcal{S} \cup \{i\}, K) - f^{3DP}(\mathbf{z}, \mathcal{S}, K) \leq f^{3DP}(\mathbf{z}, \mathcal{T} \cup \{i\}, K) - f^{3DP}(\mathbf{z}, \mathcal{T}, K). \quad (\text{A.32})$$

Note that by the construction of f^{3DP} in Problem (A.2), we have $f^{3DP}(\mathbf{z}, \mathcal{A}, K) = f^{3DP}(\mathbf{z}(\mathcal{A}), \mathcal{N}, K)$, where

$$\mathbf{z}(\mathcal{A}) \equiv \begin{cases} z_j, & \text{if } j \in \mathcal{A}, \\ 0, & \text{otherwise.} \end{cases}$$

Substituting this identity into Eq. (A.32), it suffices to prove the equivalent inequality:

$$f^{3DP}(\mathbf{z}(\mathcal{S} \cup \{i\})) - f^{3DP}(\mathbf{z}(\mathcal{S})) \leq f^{3DP}(\mathbf{z}(\mathcal{T} \cup \{i\})) - f^{3DP}(\mathbf{z}(\mathcal{T})). \quad (\text{A.33})$$

From the proof of Theorem 3, we know that f^{3DP} satisfies the *increasing differences* property:

$$f^{3DP}(\boldsymbol{\zeta} + \beta \mathbf{e}_\ell) - f^{3DP}(\boldsymbol{\zeta}) \leq f^{3DP}(\boldsymbol{\zeta} + \alpha \mathbf{e}_k + \beta \mathbf{e}_\ell) - f^{3DP}(\boldsymbol{\zeta} + \alpha \mathbf{e}_k), \quad (\text{A.34})$$

for any $\boldsymbol{\zeta} \in \mathbb{R}_+^n$ and $\alpha, \beta > 0$, where $\mathbf{e}_k$ and $\mathbf{e}_\ell$ denote the k^{th} and ℓ^{th} coordinate vectors.

We now apply this *increasing differences* property of f^{3DP} to prove Eq. (A.33) by induction. Reordering the indices so that $\mathcal{S} \setminus \mathcal{T} = \{1, \dots, L\}$ with $L = |\mathcal{S} \setminus \mathcal{T}|$, the base case follows as:

$$\begin{aligned} f^{3DP}(\mathbf{z}(\mathcal{S} \cup \{i\})) - f^{3DP}(\mathbf{z}(\mathcal{S})) &= f^{3DP}(\mathbf{z}(\mathcal{S}) + z_i \mathbf{e}_i) - f^{3DP}(\mathbf{z}(\mathcal{S})) \\ &\leq f^{3DP}(\mathbf{z}(\mathcal{S}) + z_1 \mathbf{e}_1 + z_i \mathbf{e}_i) - f^{3DP}(\mathbf{z}(\mathcal{S}) + z_1 \mathbf{e}_1), \end{aligned} \quad (\text{A.35})$$

where the equality follows from the definition of $\mathbf{z}(\mathcal{S})$, and the inequality follows by applying Eq. (A.34) with $\boldsymbol{\zeta} = \mathbf{z}(\mathcal{S})$, $\alpha = z_1$, $\beta = z_i$, and setting $k = 1$, $\ell = i$. Now, assume that for some $j' < L$, the following holds:

$$f^{3DP}(\mathbf{z}(\mathcal{S} \cup \{i\})) - f^{3DP}(\mathbf{z}(\mathcal{S})) \leq f^{3DP} \left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'} z_j \mathbf{e}_j + z_i \mathbf{e}_i \right) - f^{3DP} \left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'} z_j \mathbf{e}_j \right). \quad (\text{A.36})$$

Applying Eq. (A.34) with $\zeta = \mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'} z_j e_j$, $\alpha = z_{j'+1}$, $\beta = z_i$, and setting $k = j' + 1$, $\ell = i$, we obtain:

$$f^{3\text{DP}}\left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'} z_j e_j + z_i e_i\right) - f^{3\text{DP}}\left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'} z_j e_j\right) \leq f^{3\text{DP}}\left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'+1} z_j e_j + z_i e_i\right) - f^{3\text{DP}}\left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'+1} z_j e_j\right).$$

Combining this with Eq. (A.36), we establish:

$$f^{3\text{DP}}(\mathbf{z}(\mathcal{S} \cup \{i\})) - f^{3\text{DP}}(\mathbf{z}(\mathcal{S})) \leq f^{3\text{DP}}\left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'+1} z_j e_j + z_i e_i\right) - f^{3\text{DP}}\left(\mathbf{z}(\mathcal{S}) + \sum_{j=1}^{j'+1} z_j e_j\right).$$

Thus, by induction, Eq. (A.36) holds for all $1 \leq j' \leq L$. In particular, substituting the constructions

$$\mathbf{z}(\mathcal{T}) = \mathbf{z}(\mathcal{S}) + \sum_{j=1}^L z_j e_j, \quad \mathbf{z}(\mathcal{T} \cup \{i\}) = \mathbf{z}(\mathcal{S}) + \sum_{j=1}^L z_j e_j + z_i e_i, \quad (\text{A.37})$$

into Eq. (A.36) with $j' = L$, we establish Eq. (A.33), hence completing the proof. $\square$

Proof of Proposition 3 We prove this result by counterexample in a setting with one-to-one correspondence among primary suppliers, components, and products, and hence denote the primary sourcing cost as c_j per unit. Let $n = 3$, with $s_j \sim \text{Bernoulli}(0.9)$ (all-or-nothing disruptions) independently across suppliers, and let D follow a two-point distribution. The table below summarizes the scenarios, probabilities of D_j , and the cost parameters.

	c_j	$c_j^{3\text{DP}}$	v_j	h_j	D_j scenario 1	D_j probability 1	D_j scenario 2	D_j probability 2
$j = 1$	0.4	0.1	1	0.1	150	0.2	340	0.8
$j = 2$	0.3	0.2	0.6	0.2	140	0.1	150	0.9
$j = 3$	0.6	0.5	1	0.5	70	0.1	140	0.9

Under this setting, we have $U^{3\text{DP}}(\{1, 2, 3\}) - U^{3\text{DP}}(\{1, 2\}) = 89.47$ and $U^{3\text{DP}}(\{1, 3\}) - U^{3\text{DP}}(\{1\}) = 91.07$, violating the supermodularity condition in (5.1). Similarly, $U^{3\text{DP}}(\{1, 2, 3\}) - U^{3\text{DP}}(\{2, 3\}) = 150.44$ and $U^{3\text{DP}}(\{1, 2\}) - U^{3\text{DP}}(\{1\}) = 149.52$, contradicting submodularity. $\square$

Proof of Theorem 5 For simplicity, we will occasionally express $U^{3\text{DP}}$ and $L^{3\text{DP}}$ only as functions of K , with the understanding that $\mathcal{A}$ is fixed. Let $\bar{D}_j^* \equiv [D_j - \sum_{i \in \mathcal{S}_j} \hat{q}_{ij}^0 s_{ij}]^+$ throughout the proof.

Part i) and ii) With the convexity of $U^{3\text{DP}}$ in K established in Lemma 3, we derive a lower bound on $U^{3\text{DP}}$ as follows:

$$U^{3\text{DP}}(K) \geq U^{3\text{DP}}(0) + K\eta = \sum_{j \in \mathcal{A}} U_j^0 + K \mathbb{E}[\lambda(\mathbf{D}, \mathbf{s}, 0)] \quad (\text{A.38a})$$

$$= \sum_{j \in \mathcal{A}} U_j^0 - K \mathbb{E}_{\mathbf{D}, \mathbf{s}} \left[\max_{j \in \mathcal{A}} (v_j - c_j^{3\text{DP}}) \mathbb{I}\{\bar{D}_j^* > 0\} \right]. \quad (\text{A.38b})$$

Here, η in Eq. (A.38a) is the largest subgradient of $\kappa \mapsto U^{3\text{DP}}(\kappa)$ at $\kappa = 0$, and the inequality in Eq. (A.38a) follows from the subgradient inequality of $U^{3\text{DP}}$. Meanwhile, $\lambda(\mathbf{D}, \mathbf{s}, 0)$ in Eq. (A.38a) corresponds to the λ from Eq. (A.3), defined under $\mathbf{y} = \mathbf{D} - \hat{\mathbf{q}}^0 \circ \mathbf{s}$ and $K = 0$ for some $\mathbf{D}$ and $\mathbf{s}$, making it the largest subgradient of $\kappa \mapsto V^{3\text{DP}}(\mathbf{D} - \hat{\mathbf{q}}^0 \circ \mathbf{s}, \kappa)$ at $\kappa = 0$ by construction (see Remark 1). Applying Danskin's theorem to $U^{3\text{DP}}$, we obtain $\eta = \mathbb{E}[\lambda(\mathbf{D}, \mathbf{s}, 0)]$, validating the equality in Eq. (A.38a), while Eq. (A.38b) follows directly from the construction of $\lambda(\mathbf{D}, \mathbf{s}, 0)$ in Eq. (A.3).

Finally, defining the right-hand side of (A.38b) as $L^{3\text{DP}}(K)$ completes the proof of part (i). Part (ii) then follows taking the limit $K \rightarrow 0$ on both sides of Eq. (A.38b).

Part iii) The first term in (5.2) is trivially modular in $\mathcal{A}$, therefore to prove supermodularity of L^{3DP} in $\mathcal{A}$, it is suffice to show that the following function is submodular in $\mathcal{A}$:

$$E(\mathcal{A}) \equiv \mathbb{E}_{D,s} \left[\max_{j \in \mathcal{A}} r_j^{3DP} \mathbb{I} \{ \bar{D}_j^* > 0 \} \right] \quad (\text{A.39})$$

where $r_j^{3DP} \equiv v_j - c_j^{3DP}$. Note that:

$$E(\mathcal{A}) = \mathbb{E}_{D,s} \left[\sum_{\mathcal{A}' \subseteq \mathcal{A}} \mathbb{I} \left\{ \begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}', \\ \bar{D}_j^* = 0, \forall j \in \mathcal{A} \setminus \mathcal{A}' \end{array} \right\} \left(\max_{j \in \mathcal{A}'} r_j^{3DP} \right) \right] = \sum_{\mathcal{A}' \subseteq \mathcal{A}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}', \\ \bar{D}_j^* = 0, \forall j \in \mathcal{A} \setminus \mathcal{A}' \end{array} \right) \left(\max_{j \in \mathcal{A}'} r_j^{3DP} \right)$$

Let $\mathcal{B} \subseteq \mathcal{N}$ be an arbitrary set that $\mathcal{A} \subseteq \mathcal{B}$, and let $i \in \mathcal{N} \setminus \mathcal{B}$, first we have

$$E(\mathcal{A}) = \sum_{\mathcal{A}' \subseteq \mathcal{A}} \left[\mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}' \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in \mathcal{A} \setminus \mathcal{A}' \end{array} \right) + \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}', \\ \bar{D}_j^* = 0, \forall j \in (\mathcal{A} \setminus \mathcal{A}') \cup \{i\} \end{array} \right) \right] \left(\max_{j \in \mathcal{A}'} r_j^{3DP} \right)$$

Note that

$$\begin{aligned} E(\mathcal{A} \cup \{i\}) &= \sum_{\mathcal{A}' \subseteq \mathcal{A}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}' \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in \mathcal{A} \setminus \mathcal{A}' \end{array} \right) \max \left\{ r_i^{3DP}, \max_{j \in \mathcal{A}'} r_j^{3DP} \right\} \\ &\quad + \sum_{\mathcal{A}' \subseteq \mathcal{A}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}', \\ \bar{D}_j^* = 0, \forall j \in (\mathcal{A} \setminus \mathcal{A}') \cup \{i\} \end{array} \right) \left(\max_{j \in \mathcal{A}'} r_j^{3DP} \right) \end{aligned}$$

Hence

$$E(\mathcal{A} \cup \{i\}) - E(\mathcal{A}) = \sum_{\mathcal{A}' \subseteq \mathcal{A}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}' \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in \mathcal{A} \setminus \mathcal{A}' \end{array} \right) \max \left\{ 0, r_i^{3DP} - \max_{j \in \mathcal{A}'} r_j^{3DP} \right\} \geq 0 \quad (\text{A.40})$$

We can similarly obtain

$$E(\mathcal{B} \cup \{i\}) - E(\mathcal{B}) = \sum_{\mathcal{B}' \subseteq \mathcal{B}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{B}' \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in \mathcal{B} \setminus \mathcal{B}' \end{array} \right) \max \left\{ 0, r_i^{3DP} - \max_{j \in \mathcal{B}'} r_j^{3DP} \right\} \geq 0 \quad (\text{A.41})$$

To show the submodularity of H , we need to prove that

$$E(\mathcal{A} \cup \{i\}) - E(\mathcal{A}) \geq E(\mathcal{B} \cup \{i\}) - E(\mathcal{B})$$

Notice that from (A.40) we have

$$\begin{aligned} &E(\mathcal{A} \cup \{i\}) - E(\mathcal{A}) \\ &= \sum_{\mathcal{A}' \subseteq \mathcal{A}} \sum_{\mathcal{E} \subseteq \mathcal{B} \setminus \mathcal{A}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{A}' \cup \mathcal{E} \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in (\mathcal{A} \setminus \mathcal{A}') \cup [(\mathcal{B} \setminus \mathcal{A}) \setminus \mathcal{E}] \end{array} \right) \max \left\{ 0, r_i^{3DP} - \max_{j \in \mathcal{A}'} r_j^{3DP} \right\} \\ &= \sum_{\mathcal{B}' \subseteq \mathcal{B}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{B}' \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in \mathcal{B} \setminus \mathcal{B}' \end{array} \right) \max \left\{ 0, r_i^{3DP} - \max_{j \in \mathcal{B}' \cap \mathcal{A}} r_j^{3DP} \right\} \end{aligned}$$

where the second equality follows from the fact that any $\mathcal{B}' \subseteq \mathcal{B}$ can be uniquely decomposed into two subsets, one contained in $\mathcal{A}$ and the other in $\mathcal{B} \setminus \mathcal{A}$. Combine this result with (A.41), we obtain

$$\begin{aligned} &E(\mathcal{A} \cup \{i\}) - E(\mathcal{A}) - [E(\mathcal{B} \cup \{i\}) - E(\mathcal{B})] \\ &= \sum_{\mathcal{B}' \subseteq \mathcal{B}} \mathbb{P} \left(\begin{array}{l} \bar{D}_j^* > 0, \forall j \in \mathcal{B}' \cup \{i\}, \\ \bar{D}_j^* = 0, \forall j \in \mathcal{B} \setminus \mathcal{B}' \end{array} \right) \left(\underbrace{\max \left\{ 0, r_i^{3DP} - \max_{j \in \mathcal{B}' \cap \mathcal{A}} r_j^{3DP} \right\} - \max \left\{ 0, r_i^{3DP} - \max_{j \in \mathcal{B}'} r_j^{3DP} \right\}}_{\geq 0, \text{ since } \mathcal{B}' \cap \mathcal{A} \subseteq \mathcal{B}'} \right) \geq 0 \end{aligned}$$

Thus we have proven the submodularity of $E(\mathcal{A})$ hence the supermodularity of L^{3DP} in $\mathcal{A}$. $\square$

Proof of Proposition 4 For convenience, throughout this proof, we use the following notations:

$$r_j^{3DP} \equiv v_j - c_j^{3DP}, \quad \bar{D}_j^* \equiv \left[D_j - \sum_{i \in \mathcal{S}_j} \hat{q}_{ij}^0 s_{ij} \right]^+, \quad E(\mathcal{A}) \equiv \mathbb{E}_{D,s} \left[\max_{j \in \mathcal{A}} r_j^{3DP} \mathbb{I} \{ \bar{D}_j^* > 0 \} \right]$$

Part i) Let $\mathcal{A} \subseteq \mathcal{N}$ and $K \geq 0$ be given, then

$$E(\mathcal{A}) \leq \left(\max_{j \in \mathcal{N}} r_j^{3DP} \right) \mathbb{E}_{D,s} \left[\max_{j \in \mathcal{A}} \mathbb{I} \{ \bar{D}_j^* > 0 \} \right] = \left(\max_{j \in \mathcal{N}} r_j^{3DP} \right) \mathbb{P}(\exists j \in \mathcal{A} : \bar{D}_j^* > 0), \quad (\text{A.42})$$

which show that $\hat{L}^{3DP}(\mathcal{A}, K)$ is a lower bound of $L^{3DP}(\mathcal{A}, K)$.

Part ii) Denote the right-hand-side of (A.42) as $\hat{E}(\mathcal{A})$, then to show the supermodularity of $\hat{L}^{3DP}(\mathcal{A}, K)$, it suffices to prove that $\hat{E}(\mathcal{A})$ is submodular. Let $\mathcal{A}$ and $\mathcal{B}$ be subsets of $\mathcal{N}$ so that $\mathcal{A} \subseteq \mathcal{B}$ and let $i \in \mathcal{N} \setminus \mathcal{B}$, then

$$\begin{aligned} & \hat{E}(\mathcal{A} \cup \{i\}) - \hat{E}(\mathcal{A}) - [\hat{E}(\mathcal{B} \cup \{i\}) - \hat{E}(\mathcal{B})] \\ &= \left(\max_{j \in \mathcal{N}} r_j^{3DP} \right) \mathbb{P} \left(\{ \bar{D}_j^* = 0, \forall j \in \mathcal{A} \} \cap \{ \bar{D}_i^* = 0 \} \cap \{ \exists j \in \mathcal{B} \setminus \mathcal{A} \text{ s.t. } \bar{D}_j^* > 0 \} \right) \geq 0 \end{aligned}$$

This verifies that $\hat{E}$ is submodular, hence $\hat{L}^{3DP}$ is supermodular in $\mathcal{A}$.

Part iii) From (A.42), it is trivial to show that $L^{3DP} = \hat{L}^{3DP}$ if r_j^{3DP} are identical across $j \in \mathcal{N}$.

Part iv) For ease of notation, we define:

$$\tilde{E}(\mathcal{A}, \lambda) \equiv \frac{1}{\lambda} \log \left(\sum_{j \in \mathcal{A}} [\mathbb{P}(\bar{D}_j^* = 0) + \mathbb{P}(\bar{D}_j^* > 0) e^{\lambda r_j}] \right) \quad (\text{A.43})$$

To prove that $L^{3DP}(\mathcal{A}, K) \geq \tilde{L}^{3DP}(\mathcal{A}, K, \lambda)$ for all $\mathcal{A} \subseteq \mathcal{N}, K \geq 0$ and $\lambda > 0$, it suffices to show that $\tilde{E}(\mathcal{A}, \lambda) \geq E(\mathcal{A})$ for all $\lambda > 0$. Specifically, first note that for any collection of random variables $\{X_j : j \in \mathcal{A}\}$, it holds that:

$$\exp \left(\lambda \mathbb{E} \left[\max_{j \in \mathcal{A}} X_j \right] \right) \leq \mathbb{E} \left[\exp \left(\lambda \max_{j \in \mathcal{A}} X_j \right) \right] = \mathbb{E} \left[\max_{j \in \mathcal{A}} e^{\lambda X_j} \right] \leq \sum_{j \in \mathcal{A}} \mathbb{E} [e^{\lambda X_j}]$$

Applying logarithm for both sides gives us:

$$\mathbb{E} \left[\max_{j \in \mathcal{A}} X_j \right] \leq \frac{1}{\lambda} \log \left(\sum_{j \in \mathcal{A}} \mathbb{E} [e^{\lambda X_j}] \right) \quad (\text{A.44})$$

Substituting X_j in Eq. (A.44) with $r_j^{3DP} \mathbb{I} \{ \bar{D}_j^* > 0 \}$, the left-hand side becomes $E(\mathcal{A})$ as defined in Eq. (A.39), while the right-hand side corresponds to $\tilde{E}(\mathcal{A}, \lambda)$. Therefore, we have shown that $\tilde{E}(\mathcal{A}, \lambda) \geq E(\mathcal{A})$.

Part v) To prove that $\tilde{L}^{3DP}(\mathcal{A}, K, \lambda)$ is supermodular in $\mathcal{A}$, it suffices to show that $\tilde{E}(\mathcal{A}, \lambda)$ defined in Eq. (A.43) is submodular in $\mathcal{A}$. Note that for arbitrary $\mathcal{S} \subseteq \mathcal{N}$ and $i \notin \mathcal{S}$, it holds that:

$$\tilde{E}(\mathcal{S} \cup \{i\}, \lambda) - \tilde{E}(\mathcal{S}, \lambda) = \frac{1}{\lambda} \log \left(1 + \frac{\mathbb{P}(\bar{D}_i^* = 0) + \mathbb{P}(\bar{D}_i^* > 0) e^{\lambda r_i^{3DP}}}{\sum_{j \in \mathcal{S}} [\mathbb{P}(\bar{D}_j^* = 0) + \mathbb{P}(\bar{D}_j^* > 0) e^{\lambda r_j^{3DP}}]} \right). \quad (\text{A.45})$$

It follows that the right-hand side of Eq. (A.45) decreases with $|\mathcal{S}|$, implying that $\tilde{E}(\mathcal{A}, \lambda)$ is submodular in $\mathcal{A}$. $\square$

Proof of Theorem 6 Since $U^{3DP}(\mathcal{A}_0, K)$ and $C^{3DP}(K)$ are both convex in K , it follows that $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ is concave in K . Moreover, because $C^{3DP}(0) = 0$ and $U^{3DP}(\mathcal{A}_0, 0) = \sum_{j \in \mathcal{A}_0} U_j^0$, we have $U^{\text{tot-saved}}(\mathcal{A}_0, 0) = 0$. Therefore, there exists $K > 0$ such that $U^{\text{tot-saved}}(\mathcal{A}_0, K) > 0$ if and only if the right derivative of $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ with respect to K at $K = 0$ is positive. Indeed, if this right derivative is positive, then $U^{\text{tot-saved}}(\mathcal{A}_0, K) > 0$ for all sufficiently small $K > 0$. Conversely, if $U^{\text{tot-saved}}(\mathcal{A}_0, K) > 0$ for some $K > 0$, then by concavity,

$$\left. \frac{d^+}{d\kappa} U^{\text{tot-saved}}(\mathcal{A}_0, \kappa) \right|_{\kappa=0} \geq \frac{U^{\text{tot-saved}}(\mathcal{A}_0, K) - U^{\text{tot-saved}}(\mathcal{A}_0, 0)}{K} > 0.$$

It remains to compute the right derivative of $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ with respect to K at $K = 0$. By Theorem 5, the right derivative of $\sum_{j \in \mathcal{A}_0} U_j^0 - U^{\text{3DP}}(\mathcal{A}_0, K)$ at $K = 0$ is

$$\mathbb{E}_{\mathbf{D}, \mathbf{s}} \left[\max_{j \in \mathcal{A}_0} (v_j - c_j^{\text{3DP}}) \mathbb{I} \left\{ \left[D_j - \sum_{i \in \mathcal{S}_j} \hat{q}_{ij}^0 s_{ij} \right]^+ > 0 \right\} \right].$$

Combining this with the fact that C^{3DP} is differentiable at $K = 0$, we obtain that the right derivative of $U^{\text{tot-saved}}(\mathcal{A}_0, K)$ at $K = 0$ matches the right-hand side of Eq. (6.2). Thus, there exists $K > 0$ such that $U^{\text{tot-saved}}(\mathcal{A}_0, K) > 0$ if and only if Eq. (6.2) holds. Finally, if $U^{\text{tot-saved}}(\mathcal{A}_0, K) > 0$ for some $K > 0$, then for any $K' \in (0, K]$, concavity gives

$$U^{\text{tot-saved}}(\mathcal{A}_0, K') \geq \frac{K'}{K} U^{\text{tot-saved}}(\mathcal{A}_0, K) + \left(1 - \frac{K'}{K}\right) U^{\text{tot-saved}}(\mathcal{A}_0, 0) > 0.$$

This completes the proof. $\square$

Appendix B: Implementation Details

B.1. Benders Methods in Section 4.2

To solve the MIO (4.5) to optimality, we employ a standard Benders decomposition scheme (Birge and Louveaux 2011). Each iteration t solves a master problem to obtain a lower bound θ via a set of optimality cuts:

$$\begin{aligned} \min_{\mathbf{x} \in \{0,1\}^n, K \geq 0, \mathbf{q}, \theta} \quad & C^{\text{3DP}}(K) + \sum_{j=1}^n (1 - x_j) U_j^{\text{DB}} + \sum_{j=1}^n c_{ij} q_{ij} \mathbb{E} s_j + \theta \\ \text{s.t.} \quad & \bar{q}_{ij}^\infty x_j \leq q_{ij} \leq \bar{q}_{ij}^0 x_j, \quad \forall i \in \mathcal{S}_j, j \in \mathcal{N} \\ & \sigma^\tau + \lambda^\tau K + \mathbf{x}^\tau \boldsymbol{\alpha}^\tau + \mathbf{q}^\tau \boldsymbol{\beta}^\tau \leq \theta, \quad \forall \tau = 1 \dots t-1 \end{aligned} \quad (\text{B.1})$$

Here θ is a lower bound on the expected recourse cost. The coefficients $(\lambda^\tau, \boldsymbol{\alpha}^\tau, \boldsymbol{\beta}^\tau)$ are the subgradients of V^{3DP} (see Lemma 4) with respect to capacity K , backup selection $\mathbf{x}$, and first-stage order $\mathbf{q}$, evaluated at the solution from iteration τ , while σ^τ is the corresponding intercept. The cut generation step exploits this structure for efficient computation:

1. *Recourse Evaluation:* For each scenario, we solve the recourse problem. By Proposition 6 this reduces to a simple fractional knapsack problem and is solved efficiently with Algorithm 2.
2. *Cut Generation:* Given the optimal recourse solution, the subgradients $(\lambda, \boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma)$ admit explicit closed-form expressions derived via arguments closely analogous to those in Lemma 5.

B.2. Local Search Algorithm in Section 5.2

Under our heuristic framework (Algorithm 1), the “inner-minimization” problem from Problem (5.6) is formulated as:

$$\min_{\mathcal{A} \subseteq \mathcal{N}} \left\{ L^{\text{3DP}}(\mathcal{A}, K) + \sum_{j \in \mathcal{A}^c} (C_j^{\text{DB}} + U_j^{\text{DB}}) \right\}. \quad (\text{B.2})$$

We detail an efficient solution approach, starting by reformulating Problem (B.2) as the firm’s total profit maximization (whose objective value is reasonably assumed positive for approximation guarantees). Specifically, by reversing the sign, adding $\sum_{j \in \mathcal{N}} v_j \mathbb{E}[D_j]$, and applying the form of L^{3DP} , the objective of Problem (B.2) becomes:

$$\begin{aligned} & \sum_{j \in \mathcal{N}} (v_j \mathbb{E} D_j - C_j^{\text{DB}} - U_j^{\text{DB}}) + \sum_{j \in \mathcal{A}} (C_j^{\text{DB}} + U_j^{\text{DB}} - U_j^0) + K E(\mathcal{A}) \\ & \text{where } E(\mathcal{A}) \equiv \mathbb{E}_{\mathbf{D}, \mathbf{s}} \left[\max_{j \in \mathcal{A}} (v_j - c_j^{\text{3DP}}) \mathbb{I} \{ [D_j - \bar{q}_j^0 s_j]^+ > 0 \} \right]. \end{aligned} \quad (\text{B.3})$$

Note that $E(\mathcal{A})$ is submodular as established in Theorem 5. By defining $I \equiv \sum_{j \in \mathcal{N}} (v_j \mathbb{E} D_j - U_j^{\text{DB}} - C_j^{\text{DB}}) - C^{\text{3DP}}(K)$ and $\Delta_j \equiv U_j^{\text{DB}} + C_j^{\text{DB}} - U_j^0$, we arrive at the following reformulation of Problem (B.2), which maximizes a non-monotone, positive-valued submodular function:

$$\max_{\mathcal{A} \subseteq \mathcal{N}} I + \sum_{j \in \mathcal{A}} \Delta_j + K E(\mathcal{A}). \quad (\text{B.4})$$

The algorithm shown below is known to terminate with a solution $\hat{\mathcal{A}}$ whose objective value is at least $\frac{1}{3}$ of the optimal value of Problem (B.4) (see (Feige et al. 2011)).

Algorithm 3 Approximate Optimal Policy $\mathcal{A}$ under Fixed $K > 0$

Require: $K > 0$, $\{\Delta_j\}_{j=1}^n$, initial $\mathcal{A}_0$, and stopping criterion $\epsilon > 0$. Initialize with $t = 0$.

- 1: **while** true **do**
- 2: **if** $\exists i \in \mathcal{N} \setminus \mathcal{A}_t$ s.t. $K[E(\mathcal{A}_t \cup \{i\}) - E(\mathcal{A}_t)] + \Delta_i > \frac{\epsilon}{n^2} (I + \sum_{j \in \mathcal{A}_t} \Delta_j + E(\mathcal{A}_t)K)$ **then**
- 3: Update $\mathcal{A}_{t+1} = \mathcal{A}_t \cup \{i\}$, $t \leftarrow t + 1$, and **restart** loop.
- 4: **else if** $\exists i \in \mathcal{A}_t$ s.t. $K[E(\mathcal{A}_t \setminus \{i\}) - E(\mathcal{A}_t)] - \Delta_i > \frac{\epsilon}{n^2} (I + \sum_{j \in \mathcal{A}_t} \Delta_j + E(\mathcal{A}_t)K)$ **then**
- 5: Update $\mathcal{A}_{t+1} = \mathcal{A}_t \setminus \{i\}$, $t \leftarrow t + 1$, and **restart** loop.
- 6: **else break** (exit loop).
- 7: **end if**
- 8: **end while**
- 9: **return** The latest $\mathcal{A}_t$ at termination.

B.3. Projected SGD in Section 5.4

We present the pseudo-code for the projected SGD method outlined in Section 5.4. For generality, the algorithm we choose to present here also optimizes K for Problem (2.1). If only $U^{\text{3DP}}(\mathcal{A}, K)$ under a fixed K is of interest, this can be achieved by setting $K(t) \equiv K$ in the pseudo-code.

Throughout the algorithm, we assume access to a data sequence $\{(s^t, D^t) : t \geq 0\}$, either sampled offline from the known distribution of (s, D) or collected in real-time. Additionally, C^{3DP} is assumed convex with subdifferential ∂C^{3DP} .

Algorithm 4 Projected SGD: evaluating U^{3DP} and optimizing K under fixed $\mathcal{A}$

Require: Initialize $q(0) \geq 0$ and $K(0) \geq 0$, with step size $\rho_t > 0$ at step t . Set $t = 0$.

- 1: **while** true **do**
- 2: Compute $q^{\text{3DP}}(t)$ as the output $\bar{q}^{\text{3DP}}$ of Algorithm 2 with input $y = D^t = q(t) \circ s^t$.
- 3: Compute $\eta^y(t)$ and $\eta^K(t)$ as the η^y and η^K defined in Proposition 6 under $\bar{q}^{\text{3DP}} = q^{\text{3DP}}(t)$, $y = D^t - q(t) \circ s^t$, and $K = K(t)$.
- 4: **for** $j = 1, \dots, |\mathcal{A}|$ **do**
- 5: Update $q_{ij}(t+1) = [q_{ij}(t) - \rho_t (c_{ij} \mathbb{E}[s_{ij}] - \eta_j^y(t) s_{ij}^t)]^+$.
- 6: Update $K(t+1) = [K(t) - \rho_t (\gamma^t + g^t)]^+$ where $g^t \in \partial C^{\text{3DP}}(K(t))$ is a subgradient of C^{3DP} .
- 7: Terminate when a stopping rule is met at step t , otherwise proceed to step $t \leftarrow t + 1$.
- 8: **end while**
- 9: **return** The latest $q(t)$ and $K(t)$ at termination.

Appendix C: Numerical Experiment Details

C.1. Additional Calibration Details

C.1.1. 3DP Costs. The unit sourcing cost for product j is estimated as $\kappa_j = 0.006w_j$, where w_j is the weight in grams. This calculation assumes a plastic material cost of \$0.005/gram, marked up by 1.2 to account for additional costs.

As mentioned in the main text, we let $c^{\text{cap}} = \frac{Q^{\text{3DP}}}{M^{\text{3DP}}}$, where Q^{3DP} and M^{3DP} represent the monthly per printer depreciation cost (in \$) and the monthly material output per printer (in grams). Based on a 10-year life-span and current printer costs, we let $Q = \$41.67$. We calculate M^{3DP} as $M^{\text{3DP}} = \delta^{\text{3DP}} S^{\text{3DP}} W^{\text{3DP}} H^{\text{3DP}} T / \rho^{\text{3DP}}$, where W^{3DP} is the nozzle

width (typically 0.4 cm), H^{3DP} is the layer thickness (set to 0.01 cm), S^{3DP} is the nozzle movement speed (in cm/s), T is the total number of seconds in a month, and δ^{3DP} is the density of the non-metal printing material (typically 1 gram/cm³). This formula computes the total weight of material a printer can output during continuous operation for a month, adjusted by the infill density $\rho^{3DP} = 0.5$, to account for the fact that 3D-printed objects are typically not solid.

C.1.2. Dedicated Backup Costs. We assume the sourcing cost from a dedicated backup is $1.5\times$ the unit sourcing cost from a primary supplier, ensuring it is a) cheaper than printing and b) more expensive than primary sourcing. We similarly let C_j^{DB} be 75% of the primary sourcing cost times the expected demand. The choice of 75% reflects conversations with industry professionals that reserving capacity in high-volume, low-margin products usually entails a large retainer fee to offset the lost revenue the backup supplier could have earned by serving another firm.

C.2. Missing details for Section 7.5

For each feature embedding (marginal-only and full) under both experimental settings, we repeat the following steps 10 times and report the average out-of-sample performance:

- i) Randomly select 10% of problem instances as the test set. For “controlled correlation,” we draw 10% from the 300 base instances and apply all three variants to prevent data leakage.
- ii) Shuffle the remaining data and use the first S samples as training data.
- iii) Let S increase gradually and for each S , train CART, RBF-kernel SVM, and XGBoost with random forest using 10-fold cross-validation on the training data.
- iv) Evaluate and record out-of-sample performance on the test set.

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